Hindi Vidya Prachar Samiti's

RAMNIRANJAN JHUNJHUNWALA COLLEGE OF ARTS , SCIENCE AND COMMERCE

(EMPOWERED AUTONOMOUS)

Advanced Data Analytics

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Class : Msc. Data Science and Artificial Intelligence Part II

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CERTIFICATE

This is to Hitesh Ram of Msc. Data Science and Artificial Intelligence Roll No-10405 has successfully completed the practical of Advanced Data Analytics during the Academic Year 2025-2026.

Date :

(Prof. Neha Ansari)
Prof-In-Charge

External Examiner

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Practical 1

Learn all the basics of R-Programming (Data types ,Variables Operators etc.)

Sample data frame

```
> students <- data.frame(
+   Name = c("Alice","Bob","Charlie","Alice","Bob","Charlie"),
+   Subject = c("Math","Math","Math","Science","Science","Science"),
+   Score = c(90,85,88,92,81,95),
+   Age = c(14,15,14,14,15,14)
+ )
```

way 1

Mean Score by Name

```
> aggregate(Score~Name, data=students, FUN=mean)
     Name Score
1   Alice  91.0
2     Bob  83.0
3 Charlie  91.5
```

Mean score by Subject and Name

```
> aggregate(Score~Subject + Name,data=students,FUN=mean)
  Subject    Name Score
1    Math   Alice    90
2 Science   Alice    92
3    Math     Bob    85
4 Science     Bob    81
5    Math Charlie    88
6 Science Charlie    95
```

way 2

Average score per subject

```
> tapply(students$Score,students$Subject,mean)
    Math  Science
87.66667 89.33333
```

Score by Subject and Name

```
> tapply(students$Score,list(students$Name,students$Subject),mean)
        Math Science
Alice     90      92
Bob       85      81
Charlie   88      95
```

way 3

Group wise summary

```
> by(students$Score, students$Subject,summary)
students$Subject: Math
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  85.00   86.50   88.00   87.67   89.00   90.00
---------------------------------------------------------------
students$Subject: Science
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  81.00   86.50   92.00   89.33   93.50   95.00
```

by subject and name

```
> by(students$Score,list(students$Name,students$Subject),mean)
: Alice
: Math
[1] 90
-------------------------------------------------------------
: Bob
: Math
[1] 85
-------------------------------------------------------------
: Charlie
: Math
[1] 88
-------------------------------------------------------------
: Alice
: Science
[1] 92
-------------------------------------------------------------
: Bob
: Science
[1] 81
-------------------------------------------------------------
: Charlie
: Science
[1] 95
```

way 4

```
> library('dplyr')
```

Average score by Name

```
> students %>%
+   group_by(Name) %>%
+   summarise(Avg_Score=mean(Score))
# A tibble: 3 × 2
  Name    Avg_Score
  <chr>       <dbl>
1 Alice        91
2 Bob          83
3 Charlie      91.5
```

Multiple grouping

```
> students %>%
+   group_by(Name,Subject) %>%
+   summarise(Avg = mean(Score),.groups = "drop")
# A tibble: 6 × 3
  Name    Subject   Avg
  <chr>   <chr>   <dbl>
1 Alice   Math       90
2 Alice   Science    92
3 Bob     Math       85
4 Bob     Science    81
5 Charlie Math       88
6 Charlie Science    95
```

way 5

```
install.packages("data.table")
library(data.table)
dt<- data.table(students)
```

Mean score by name

```
> dt[,.(Avg_Score = mean(Score)), by=Name]
      Name Avg_Score
    <char>     <num>
1:   Alice      91.0
2:     Bob      83.0
3: Charlie      91.5
```

Mean and sum by subject

```
> dt[,.(Avg_Score= mean(Score),total=sum(Score)),by=Subject]
   Subject Avg_Score total
    <char>     <num> <num>
1:    Math  87.66667   263
2: Science  89.33333   268
```

Group by two variables

```
> dt[,.(Avg =mean(Score)),by =.(Name,Subject)]
      Name Subject   Avg
    <char>  <char> <num>
1:   Alice    Math    90
2:     Bob    Math    85
3: Charlie    Math    88
4:   Alice Science    92
5:     Bob Science    81
6: Charlie Science    95
```

```
> students %>%
+   count(Name) #same as group_by(Name) %>% summarise(n=n())
     Name n
1   Alice 2
2     Bob 2
3 Charlie 2
```

```
install.packages("jsonlite")
library(jsonlite)
```

Create a sample data frame

```
> stud <- data.frame(
+   name = c('alice','bob','marty'),
+   age = c(15,12,14),
+   score = c(90,85,88)
+ )
```

convert to json

```
> stud_json <- toJSON(stud,pretty =TRUE)
> stud_json
[
  {
    "name": "alice",
    "age": 15,
    "score": 90
  },
  {
    "name": "bob",
    "age": 12,
    "score": 85
  },
  {
    "name": "marty",
    "age": 14,
    "score": 88
  }
]
```

write json to file

```
> write(stud_json,file='stud.json')
> getwd()
[1] "C:/Users/sudha/Documents"
```

Read JSON file

```
> stud_data <- fromJSON("stud.json")
> print(stud_data)
   name age score
1 alice  15    90
2   bob  12    85
3 marty  14    88
```

csv file

```
> stud <- data.frame(
+   name = c('alice','bob','marty'),
+   age = c(15,12,14),
+   score = c(90,85,88)
+ )
> stud
   name age score
1 alice  15    90
2   bob  12    85
3 marty  14    88
```

```
> write.csv(stud,"stud.csv", row.names=FALSE)
> stud_csv<- read.csv("stud.csv")
> print(stud_csv)
   name age score
1 alice  15    90
2   bob  12    85
3 marty  14    88
```

Read a tsv file using read.table()

```
> data <- read.table("stud.tsv",sep="\t",header=TRUE)
> print(data)
   name age score
1 alice  15    90
2   bob  12    85
3 marty  14    88
```

```
install.packages("readxl")
install.packages("writexl")
```

```
> library(readxl)
> library(writexl)
> write_xlsx(stud,"stud.xlsx")
> data_excel<-read_excel("stud.xlsx")
> print(data_excel)
# A tibble: 3 × 3
  name    age score
  <chr> <dbl> <dbl>
1 alice    15    90
2 bob      12    85
3 marty    14    88
```

```
> prof <- data.frame(
+   name = c('alice','bob','marty'),
+   age = c(15,12,14),
+   score = c(90,85,88)
+ )
> write_xlsx(list(Sheet1= stud,Sheet2=prof),"multi_sheet.xlsx")
> read_xlsx("multi_sheet.xlsx")
# A tibble: 3 × 3
  name    age score
  <chr> <dbl> <dbl>
1 alice    15    90
2 bob      12    85
3 marty    14    88
```

Practical 2

Implement R loops with different examples

```
> install.packages("dplyr")
> library(dplyr)
```

Example 1: Using For loop:

```
> n <- 5
> fact <-1
> for (i in 1:n){
+   fact <- fact * i
+ }
> print(paste("Factorial of ", n,  "is", fact))
[1] "Factorial of  5 is 120"
```

Example 2: Using While loop:

```
> n <- 5
> i <- 1
> while(i <= n){
+   fact <- fact * 1
+   i <- i + 1
+ }
> print(paste("Factorial of", n, "is",fact))
[1] "Factorial of 5 is 120"
```

Example 3: Using repeat loop: [Ask for number > 10]

```
> inputs <- c(3,7,10,12)
> i <- 1
> repeat {
+   num <- inputs[i]
+   print(paste("You entered:", num))
+ if (num > 10) {
+   print("Thank you!")
+   break
+  }
+ i <- i + 1
+ }
[1] "You entered: 3"
[1] "You entered: 7"
[1] "You entered: 10"
[1] "You entered: 12"
[1] "Thank you!"
```

Example 4: Using While loop:

```
> inputs <- c(2,5,9,15)
> i <- 1
> num <- inputs[i]
> while (num <= 10) {
+   print(paste("You entered:", num))
+   i <- i + 1
+   num <- inputs[i]
+ }
[1] "You entered: 2"
[1] "You entered: 5"
[1] "You entered: 9"
> print("Thank you!")
[1] "Thank you!"
```

Example 5: Print triangle pattern

Using For loop:

```
> rows <- 4
> for (i in 1:rows){
+   line <- rep("*", i)
+   cat(paste(line, collapse=" "),  "\n")
+ }
*
* *
* * *
* * * *
```

Using While loop:

```
> rows <- 4
> i <- 1
> while ( i <= rows){
+   line <- rep("*", i)
+   cat(paste(line,collapse = " "), "\n")
+   i <- i + 1
+ }
*
* *
* * *
* * * *
```

Example 6: Print student with score 80

```
> df <- data.frame(Name=c("Alice","Bob","Charlie"), Score=c(85,90,78))
> #using for loop:
> for (i in 1:nrow(df)) {
+   if (df$Score[i] > 80) print(df$Name[i])
+ }
[1] "Alice"
[1] "Bob"
> #Using while loop
> i <- 1
> while (i <= nrow(df)) {
+   if (df$Score[i] > 80) print(df$Name[i])
+   i <- i + 1
+ }
[1] "Alice"
[1] "Bob"
```

Example 7: Add 5 bonus marks

Using for loop:

```
> for (i in 1:nrow(df)) {
+   df$Score[i] <- df$Score[i] + 5
+ }
> print(df)
     Name Score
1   Alice    90
2     Bob    95
3 Charlie    83
```

Using While loop:

```
> i <- 1
> while (i <= nrow(df)) {
+   df$Score[i] <- df$Score[i] + 5
+   i <- i + 1
+ }
> print(df)
     Name Score
1   Alice    95
2     Bob   100
3 Charlie    88
```

Example 8: Add pass/fail result

Using for loop:

```
> df$Result <- NA
> for (i in 1:nrow(df)) {
+   if (df$Score[i] >= 50) df$Result[i] <- "Pass" else df$Result[i] <- "Fail"
+ }
> print(df)
     Name Score Result
1   Alice    95   Pass
2     Bob   100   Pass
3 Charlie    88   Pass
```

Using While loop:

```
> df$Result <- NA
> i <- 1
> while (i <= nrow(df)) {
+   df$Result[i] <- ifelse(df$Score[i] >= 50, "Pass", "Fail")
+   i <- i + 1
+ }
> print(df)
     Name Score Result
1   Alice    95   Pass
2     Bob   100   Pass
3 Charlie    88   Pass
```

Example 9: sum list items using lapply

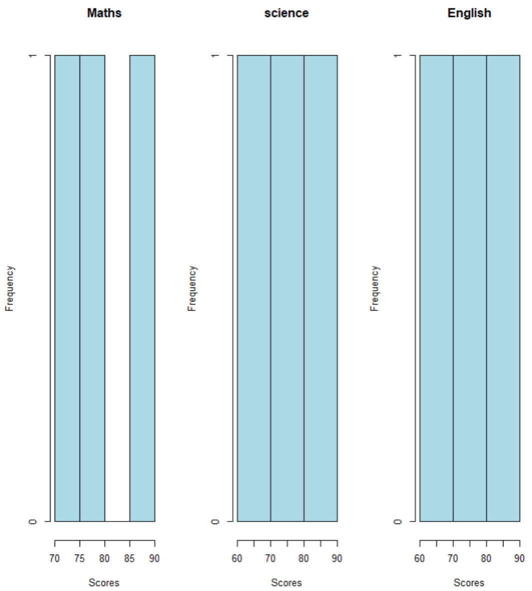
```
> my_list  <- list(x = 1:4, y=5:7,z=8:10)
> lapply(my_list,sum)
$x
[1] 10

$y
[1] 18

$z
[1] 27
```

Example 10: max of each column using sapply

```
> df <- data.frame(Maths=c(80,90,70), science=c(75,85,60),English=c(65,88,72))
> sapply(df,max)
  Maths science English
     90      85      88
```

Example 11: Histogram using loop

```
> df <- data.frame(Maths=c(80,90,70), science=c(75,85,60),English=c(65,88,72))
> par(mfrow=c(1,3))
> for (col in names(df))
+ {
+   hist(df[[col]],main=col,xlab='Scores', col='lightblue')
+ }
```



Example 12 : Applying Normalization using apply function

```
> normalize <- function(x) (x-min(x)) / (max(x)- min(x))
> df_norm <- as.data.frame(apply(df,2,normalize))
> print(df_norm)
  Maths science   English
1   0.5     0.6 0.0000000
2   1.0     1.0 1.0000000
3   0.0     0.0 0.3043478
```

Example 13 : Add Grade column to the data frame and assign value to it

```
> df <- data.frame(Maths=c(80,90,70), science=c(75,85,60),English=c(65,88,72))
> df$Grade<-NA
> print(df)
  Maths science English Grade
1    80      75      65    NA
2    90      85      88    NA
3    70      60      72    NA
> for (i in 1:nrow(df)) {
+ avg <- mean(as.numeric(df[i,1:3]))
+ print(avg)
+ }
[1] 73.33333
[1] 87.66667
[1] 67.33333
> for (i in 1:nrow(df)) {
+ avg <- mean(as.numeric(df[i,1:3]))
+ if (avg >= 85)
+   df$Grade[i] <- "A"
+ else if (avg >= 70)
+   df$Grade[i] <- "B"
+ else
+   df$Grade[i] <- "c"
+ }
> print(df)
  Maths science English Grade
1    80      75      65     B
2    90      85      88     A
3    70      60      72     c
```

Practical 3

Learn the basic of functions in R and Implement them with examples

1. **Function without arguments**

```
hello_function <- function(){
  print("Hello, R")
}
hello_function()
l] "Hello, R"
```

2. **Function with arguments**

```
add_number <- function(a,b)
{
  sum <- a + b
  return(sum)
}
add_number(10,5)
l] 15
```

3. **Function with default arguments**

```
> power_func <- function(x,power=2){
+   return(x^power)
+ }
> power_func(4)
[1] 16
> power_func(4,3)
[1] 64
```

4. **Function returning multiple values**

```
> math_ops <- function(a,b)
+ {
+   add= a+b
+   sub= a-b
+   mul= a*b
+   div= a/b
+   return(c(add,sub,mul,div))
+ }
> math_ops(5,4)
[1]  9.00  1.00 20.00  1.25
```

5. **Anonymous (inline) functions**

```
> sapply(1:5, function(x) x^2)
[1]  1  4  9 16 25
```

6. **Functions with variable arguments**

```
> sum_all <- function(...)
+ {
+   return(sum(...))
+ }
> sum_all(1,2,3)
[1] 6
> sum_all(1,2,3,4,5)
[1] 15
```

Practical 4

Implement Data frames in R. Write a program to join columns and rows in a data frame using c bind() and r bind() in R

Demonstration of c bind(), r bind() in data frame

```
> Name <- c("abc","def","ghi")
> Age <- c(25,30,35)
> City <- c("New York", "Los Angeles","Chicago")
> df <- data.frame(Name,Age,City)
> df
  Name Age        City
1  abc  25    New York
2  def  30 Los Angeles
3  ghi  35     Chicago
> str(df)
'data.frame':   3 obs. of  3 variables:
 $ Name: chr  "abc" "def" "ghi"
 $ Age : num  25 30 35
 $ City: chr  "New York" "Los Angeles" "Chicago"
> dim(df)
[1] 3 3
> summary(df)
     Name                Age            City          
 Length:3           Min.   :25.0   Length:3          
 Class :character   1st Qu.:27.5   Class :character  
 Mode  :character   Median :30.0   Mode  :character  
                    Mean   :30.0                     
                    3rd Qu.:32.5                     
                    Max.   :35.0                     
> Gender <- c("F","M","M")
> Salary <- c(50000,15000,20000)
> df <- cbind(df,Gender, Salary)
> print(df)
  Name Age        City Gender Salary
1  abc  25    New York      F  50000
2  def  30 Los Angeles      M  15000
3  ghi  35     Chicago      M  20000
> new_row <- data.frame(Name='Louis', Age=35, City="New York", Gender='M',Salary=45000)
> df <- rbind(df,new_row)
> print(df)
   Name Age        City Gender Salary
1   abc  25    New York      F  50000
2   def  30 Los Angeles      M  15000
3   ghi  35     Chicago      M  20000
4 Louis  35    New York      M  45000
> new_rows <- data.frame(Name=c('William','Diana'), Age=c(40,36),
+                        City=c("Paris","London"), Gender=c('M','F'),
+                        Salary=c(45000,30000))
> df <- rbind(df,new_rows)
> print(df)
     Name Age        City Gender Salary
1     abc  25    New York      F  50000
2     def  30 Los Angeles      M  15000
3     ghi  35     Chicago      M  20000
4   Louis  35    New York      M  45000
5 William  40       Paris      M  45000
6   Diana  36      London      F  30000
7 William  40       Paris      M  45000
8   Diana  36      London      F  30000
```

Practical 5

Implement different String Manipulation functions in R

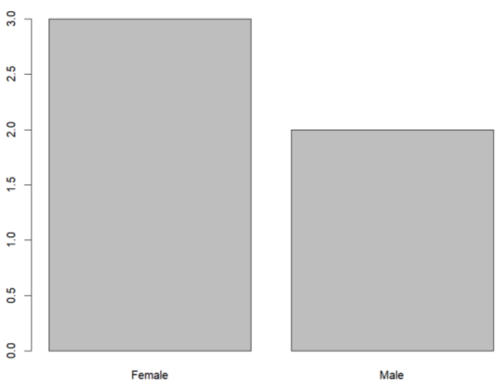
```
> str1
[1] "Hello R Programming"
> str2 <- "Data Science with R "
> str2
[1] "Data Science with R "
> cat("Upper Case:", toupper(str1))
Upper Case: HELLO R PROGRAMMING
> cat("Lower Case:", tolower(str1))
Lower Case: hello r programming
> cat("Trimmed:", trimws(str2))
Trimmed: Data Science with R
> cat("No of character",nchar(str1))
No of character 19
> cat("Substring:",substring(str1,7,))
Substring: R Programming
> cat("Substring:",substring(str1,7,9))
Substring: R P
> paste("R","Programming",sep=',')
[1] "R,Programming"
> paste("R","Programming", sep="")
[1] "RProgramming"
> paste("R","Programming", sep="-")
[1] "R-Programming"
> paste0("R","Programming")
[1] "RProgramming"
> strsplit("apple,banana,mango",",")
[[1]]
[1] "apple"  "banana" "mango"
> strsplit(str1," ")
[[1]]
[1] "Hello"       "R"           "Programming"

> sub("R","r",str1)
[1] "Hello r Programming"
> sub("R","python",str1)
[1] "Hello python Programming"
> grep("R",str1)
[1] 1
> words <- c("apple","banana","grape","apricot")
> words
[1] "apple"   "banana"  "grape"   "apricot"
> grep("^a", words, value=TRUE)
[1] "apple"   "apricot"
> grep("a$", words, value=TRUE)
[1] "banana"
> words <- c("apple","bananA","grape","apricot")
> grep("a$",words,value=TRUE)
character(0)
> grep("a$", words,value=TRUE,ignore.case=TRUE)
[1] "bananA"
> str = " Data Science with R "
> cat(trimws(str))
Data Science with R
> str = " Data Science with R "
> cat(trimws(str,which="right"))
 Data Science with R
> str =" Data Science with R "
> cat(trimws(str,which="left"))
Data Science with R
```

Practical 6

Implement different data structures in R (Vectors, Lists, Data Frames)

Factor and dataframe

```
install.packages('dplyr')
library(dplyr)
> gender <- factor(c('Male', 'Female', 'Male', 'Female', 'Female'))
> print(gender)
[1] Male   Female Male   Female Female
Levels: Female Male
> levels(gender)
[1] "Female" "Male"
> nlevels(gender)
[1] 2
> grades <- factor(c('B', 'A', 'C', 'A', 'B'),
+                  levels = c('C', 'B', 'A'),
+                  ordered = TRUE)
> print(grades)
[1] B A C A B
Levels: C < B < A
> grades[1] > grades[3]
[1] TRUE
> grades[1]
[1] B
Levels: C < B < A
> grades[3]
[1] C
Levels: C < B < A
> grades[2]
[1] A
Levels: C < B < A
> city <- c('Delhi', 'Mumbai', 'Delhi', 'Kolkata')
> city
[1] "Delhi"   "Mumbai"  "Delhi"   "Kolkata"
> city_factor = factor(city)
> levels(city_factor)
[1] "Delhi"   "Kolkata" "Mumbai"
> df <- data.frame(
+   Name = c('Aman', 'Sara', 'Ravi'),
+   Gender = factor(c('Male', 'Female', 'Male')),
+   Grade = factor(c('A', 'B', 'A'), ordered = TRUE, levels = c('C', 'B', 'A'))
+ )
> df
  Name Gender Grade
1 Aman   Male     A
2 Sara Female     B
3 Ravi   Male     A
> barplot(table(gender))
```



```
> array_data <- array(1:8, dim = c(2,2,2))
> print(array_data)
, , 1

     [,1] [,2]
[1,]    1    3
[2,]    2    4

, , 2

     [,1] [,2]
[1,]    5    7
[2,]    6    8

> array_data <- array(1:8, dim = c(2,4))
> print(array_data)
     [,1] [,2] [,3] [,4]
[1,]    1    3    5    7
[2,]    2    4    6    8
> vector <- c(1,2,3)
> vector
[1] 1 2 3
> m = matrix(1:6, nrow = 3)
> m
     [,1] [,2]
[1,]    1    4
[2,]    2    5
[3,]    3    6
> l = list(1,2,'Raj', TRUE)
> l
[[1]]
[1] 1

[[2]]
[1] 2

[[3]]
[1] "Raj"

[[4]]
[1] TRUE
> l2 = list(list(1,2,'Raj', TRUE),
+           'a', TRUE)
> l2
[[1]]
[[1]][[1]]
[1] 1

[[1]][[2]]
[1] 2

[[1]][[3]]
[1] "Raj"

[[1]][[4]]
[1] TRUE


[[2]]
[1] "a"

[[3]]
[1] TRUE
```

```
> temperature = factor(c('High', 'Low', 'Medium', 'Low', 'High'),
+                      ordered = TRUE,
+                      levels = c('Low', 'Medium', 'High'))
> print(temperature)
[1] High   Low    Medium Low    High
Levels: Low < Medium < High
> levels(temperature)
[1] "Low"    "Medium" "High"
> nlevels(temperature)
[1] 3
> install.packages('tidyr')
> library(tidyr)
> data_wide <- data.frame(
+   Name = c('A', 'B'),
+   Math = c(90, 80),
+   Science = c(85, 75)
+ )
> data_long <- pivot_longer(data_wide, cols = c(Math, Science),
+                           names_to = 'Subject', values_to = 'Score')
> data_wide
  Name Math Science
1    A   90      85
2    B   80      75
> data_long
# A tibble: 4 × 3
  Name  Subject Score
  <chr> <chr>   <dbl>
1 A     Math       90
2 A     Science    85
3 B     Math       80
4 B     Science    75
> data_long2 = data.frame(Name = c('A', 'A', 'B', 'B'),
+                         Subject = c('Math', 'Science', 'Math', 'Science'),
+                         Score = c(90, 85, 80, 75))
> data_wide2 = pivot_wider(data_long2, names_from = Subject, values_from = Score)
> data_long2
  Name Subject Score
1    A    Math    90
2    A Science    85
3    B    Math    80
4    B Science    75
> data_wide2
# A tibble: 2 × 3
  Name   Math Science
  <chr> <dbl>   <dbl>
1 A        90      85
2 B        80      75
```

Date

```
> date_str <- "2025-07-08"
> date_obj <- as.Date(date_str)
> print(date_str)
[1] "2025-07-08"
> date_str <- "08/07/2025"
> date_obj <- as.Date(date_str, format = '%d/%m/%Y')
> final_date = format(date_obj,"%d/%m/%Y")
> print(date_str)
[1] "08/07/2025"
> today <- Sys.Date()
> formatted <- format(today, "%B %d, %Y")
> print(formatted)
[1] "September 05, 2025"
> date_val <- as.Date("2025-07-08")
> year <- format(date_val,"%Y")
> month <- format(date_val,"%m")
> day <- format(date_val,"%d")
> cat("Year:",year,"Month:",month,"Day:",day)
Year: 2025 Month: 07 Day: 08
> start_date <- as.Date("2025-07-01")
> end_date <- as.Date("2025-07-08")
> date_seq <- seq(start_date,end_date, by="day")
> print(date_seq)
[1] "2025-07-01" "2025-07-02" "2025-07-03" "2025-07-04" "2025-07-05" "2025-07-06"
[7] "2025-07-07" "2025-07-08"
```

Character Manipulation in R

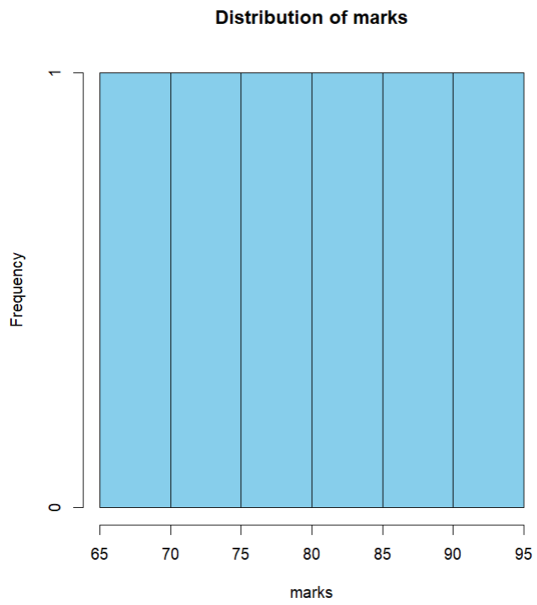
```
> text <- "Hello R"
> nchar(text)
[1] 7
> toupper("r language")
[1] "R LANGUAGE"
> tolower("HELLO")
[1] "hello"
> text <- "abcdef"
> substr(text,2,4)
[1] "bcd"
> text <- "apple banana apple"
> sub("apple","orange",text)
[1] "orange banana apple"
> gsub("apple","orange",text)
[1] "orange banana orange"
> paste("R","Language")
[1] "R Language"
> paste0("R", "Lang")
[1] "RLang"
> paste("R","Language",sep=';')
[1] "R;Language"
```

Practical 7

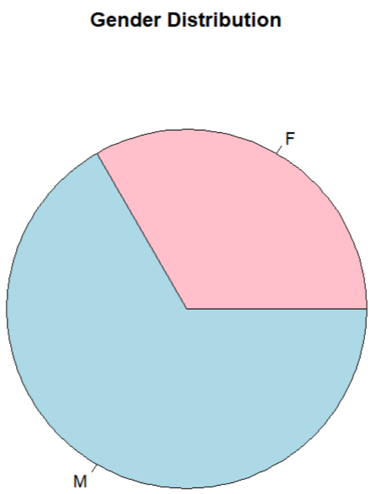
Write a program to read a csv file and perform basic analysis in R

```
> Name <- c("Alice","Bob","Charlie","Devid","Eva","Frank")
> Age <- c(20,22,21,23,22,24)
> Gender <- c("F","M","M","M", "F","M")
> Marks <- c(85,78,92,67,88,74)
> df <- data.frame(Name,Age,Gender,Marks)
> df
     Name Age Gender Marks
1   Alice  20      F    85
2     Bob  22      M    78
3 Charlie  21      M    92
4   Devid  23      M    67
5     Eva  22      F    88
6   Frank  24      M    74
> write.csv(df, "Students.csv", row.names = FALSE)
> students <- read.csv("Students.csv", header = TRUE, stringsAsFactors = FALSE)
> print("First 6 rows: ")
[1] "First 6 rows: "
> print(head(students))
     Name Age Gender Marks
1   Alice  20      F    85
2     Bob  22      M    78
3 Charlie  21      M    92
4   Devid  23      M    67
5     Eva  22      F    88
6   Frank  24      M    74
> print("Summary of Data:")
[1] "Summary of Data:"
> print(summary(students))
     Name                Age           Gender              Marks
 Length:6           Min.   :20.00   Length:6           Min.   :67.00
 Class :character   1st Qu.:21.25   Class :character   1st Qu.:75.00
 Mode  :character   Median :22.00   Mode  :character   Median :81.50
                    Mean   :22.00                      Mean   :80.67
                    3rd Qu.:22.75                      3rd Qu.:87.25
                    Max.   :24.00                      Max.   :92.00
> print("Structure of Data Frame:")
[1] "Structure of Data Frame:"
> print(str(students))
'data.frame':   6 obs. of  4 variables:
 $ Name  : chr  "Alice" "Bob" "Charlie" "Devid" ...
 $ Age   : int  20 22 21 23 22 24
 $ Gender: chr  "F" "M" "M" "M" ...
 $ Marks : int  85 78 92 67 88 74
NULL
> cat("Number of students:", nrow(students),"\n")
Number of students: 6
> cat("Avg of age:",mean(students$Age))
Avg of age: 22
> cat("Avg marks:", mean(students$Marks))
Avg marks: 80.66667
> cat("Maximum Marks:", max(students$Marks))
Maximum Marks: 92
> cat("Minimum Marks:",min(students$Marks))
Minimum Marks: 67
> avg_marks_by_gender <- aggregate(students$Marks, by =list(df$Gender), FUN = mean)
> print(avg_marks_by_gender)
  Group.1     x
1       F 86.50
2       M 77.75
```

```
> print("Average Marks by gender:")
[1] "Average Marks by gender:"
> print(avg_marks_by_gender)
  Group.1     x
1       F 86.50
2       M 77.75
> hist(students$Marks,
+       main ="Distribution of marks",
+       xlab='marks',
+       col='skyblue',
+       border='black')
```




```
> gender_counts <- table(students$Gender)
> print(gender_counts)

F M
2 4
> pie(gender_counts,
+      labels=names(gender_counts),
+      main="Gender Distribution",
+      col=c('pink','lightblue'))
```

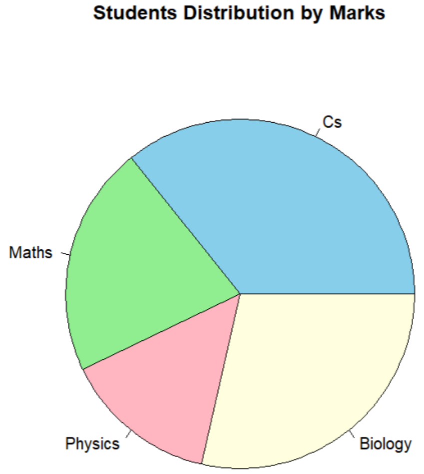


```
> library('dplyr')
> sample(students,3)
  Age Gender    Name
1  20      F   Alice
2  22      M     Bob
3  21      M Charlie
4  23      M   Devid
5  22      F     Eva
6  24      M   Frank
> sample_n(students,2)
     Name Age Gender Marks
1     Eva  22      F    88
2 Charlie  21      M    92
> sample_frac(students,0.3)
   Name Age Gender Marks
1 Devid  23      M    67
2   Bob  22      M    78
```

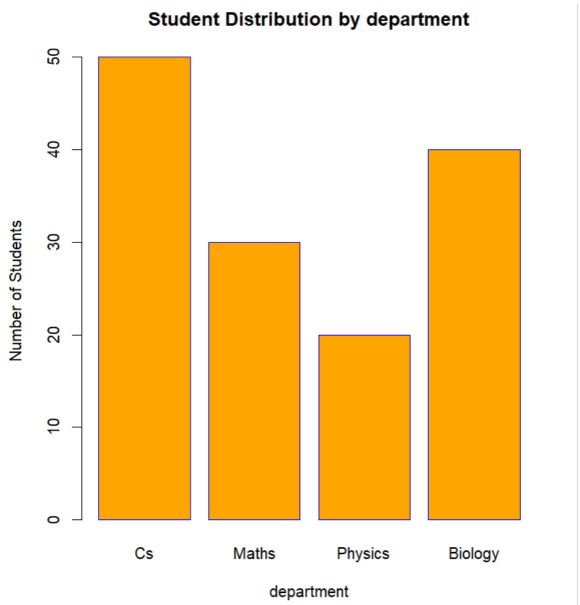
Practical 8

Create pie charts and bar charts using R

```
> departments <- c("Cs","Maths","Physics","Biology")
> students <- c(50,30,20,40)
> pie(students,
+     labels=departments,
+     main="Students Distribution by Marks",
+     col=c("skyblue","lightgreen","lightpink","lightyellow"))
```

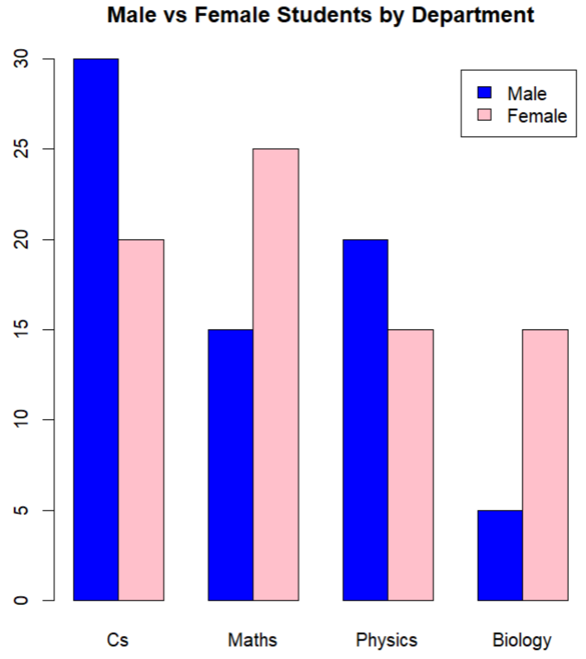



```
> barplot(students,
+         names.arg= departments,
+         main = "Student Distribution by department",
+         xlab = 'department',
+         ylab= "Number of Students",
+         col='orange',
+         border='blue')
```




```
> # Data for male & female students
> students_data <- matrix(c(30,20,15,25,20,15,5,15), nrow= 2)
> rownames(students_data) <- c("Male","Female")
> colnames(students_data) <- departments
> print(students_data)
       Cs Maths Physics Biology
Male   30    15      20       5
Female 20    25      15      15
```

```
> #side-by-side bar Chart
> barplot(students_data,
+         beside= TRUE,
+         main = "Male vs Female Students by Department",
+         col = c("blue","pink"),
+         legend = rownames(students_data))
```



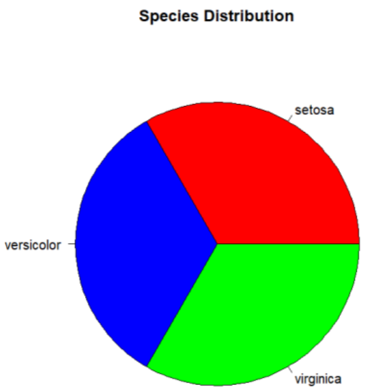
Practical 9

Create a dataset and do statistical analysis on the data

```
> data(iris)
> head(iris)
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
> str(iris)
'data.frame':   150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
> summary(iris)
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width          Species  
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100   setosa    :50  
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300   versicolor:50  
 Median :5.800   Median :3.000   Median :4.350   Median :1.300   virginica :50  
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199                  
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800                  
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500                  
```

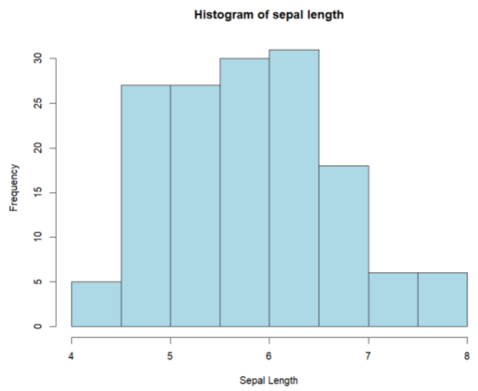
pie chat of species ditribution

```
> pie(table(iris$Species),col=c("red","blue","green"),
+     main = "Species Distribution")
```



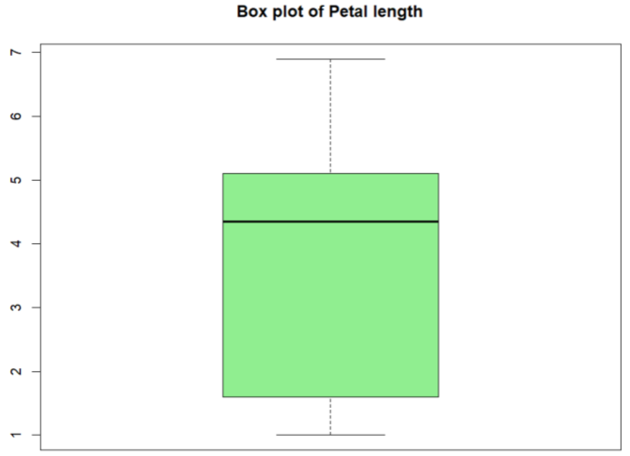
Histogram of sepal length

```
> hist(iris$Sepal.Length,col="lightblue",
+     main ="Histogram of sepal length",
+     xlab  = "Sepal Length")
```



Boxplot of Petal Length

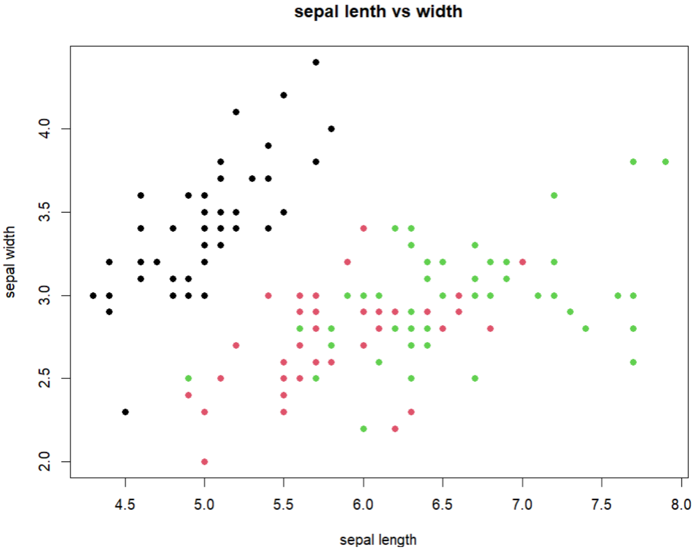
```
> boxplot(iris$Petal.Length,col="lightgreen",
+         main='Box plot of Petal length')
```




```
> #Frequency table of species
> table(iris$Species)

    setosa versicolor  virginica
        50         50         50
```

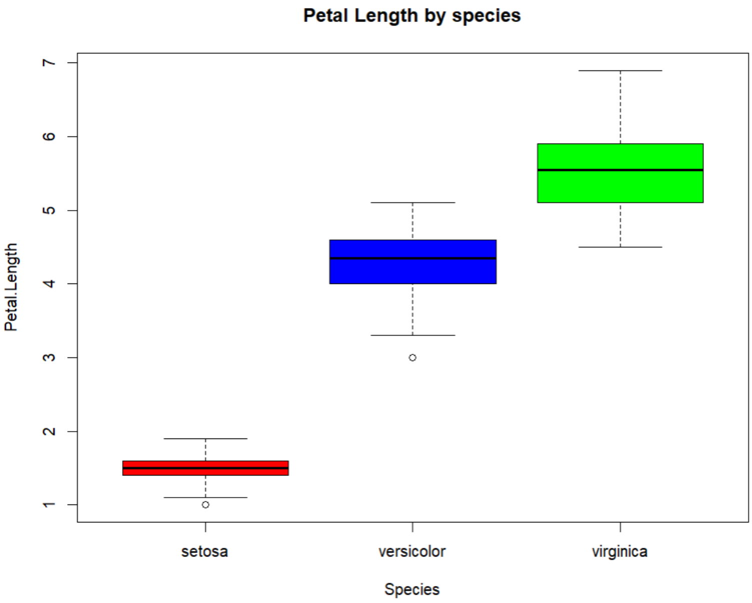
scatter plot Sepal.Length vs Sepal.Length

```
> plot(iris$Sepal.Length,iris$Sepal.Width,
+      col= iris$Species,pch=19,
+      main ="sepal lenth vs width",
+      xlab= "sepal length", ylab="sepal width")
```



Boxplot of petal.length by species

```
> boxplot(Petal.Length ~ Species,data=iris,
+         main="Petal Length by species",
+         col=c("red","blue","green"))
```

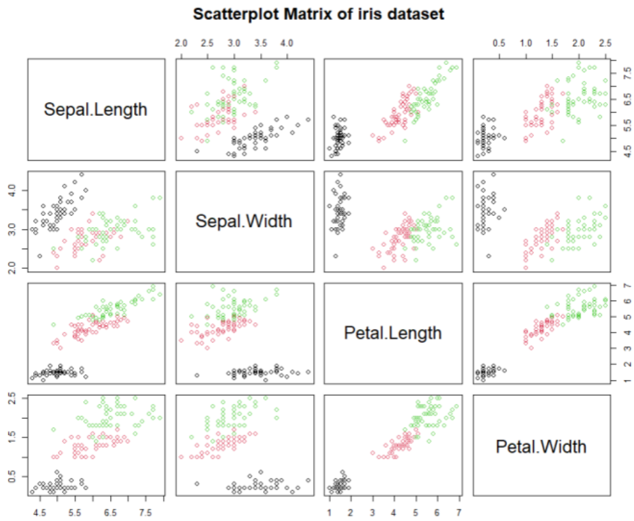


correlation between numeric variables

```
> cor(iris[,1:4])
             Sepal.Length Sepal.Width Petal.Length Petal.Width
Sepal.Length    1.0000000  -0.1175698    0.8717538   0.8179411
Sepal.Width    -0.1175698   1.0000000   -0.4284401  -0.3661259
Petal.Length    0.8717538  -0.4284401    1.0000000   0.9628654
Petal.Width     0.8179411  -0.3661259    0.9628654   1.0000000
```

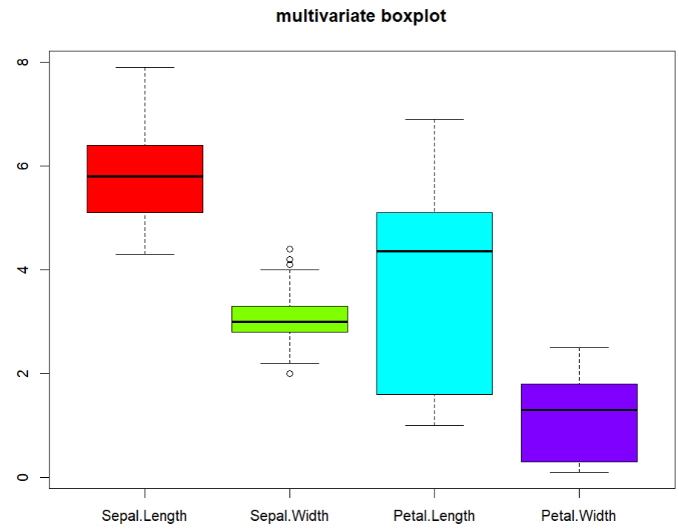
Pairplot (scatter matrix)

```
> pairs(iris[,1:4], col=iris$Species,
+       main="Scatterplot Matrix of iris dataset")
```



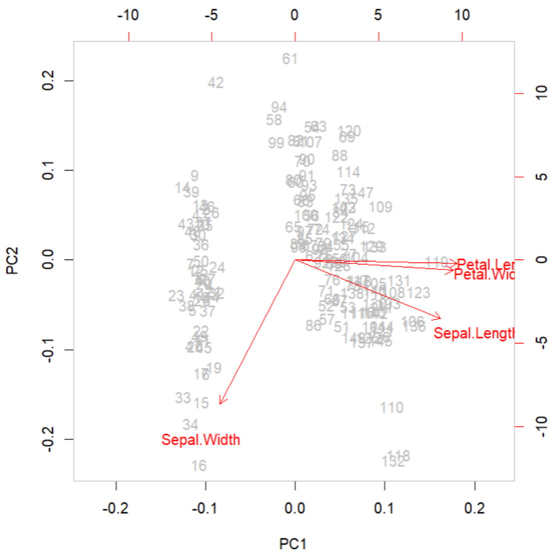
Muti-variate boxplot

```
> boxplot(iris[,1:4],main="multivariate boxplot",
+         col=rainbow(4))
```



Principal Component Analysis (PCA)

```
> pca <- prcomp(iris[,1:4], scale.=TRUE)
> biplot(pca,col=c("gray","red"))
```

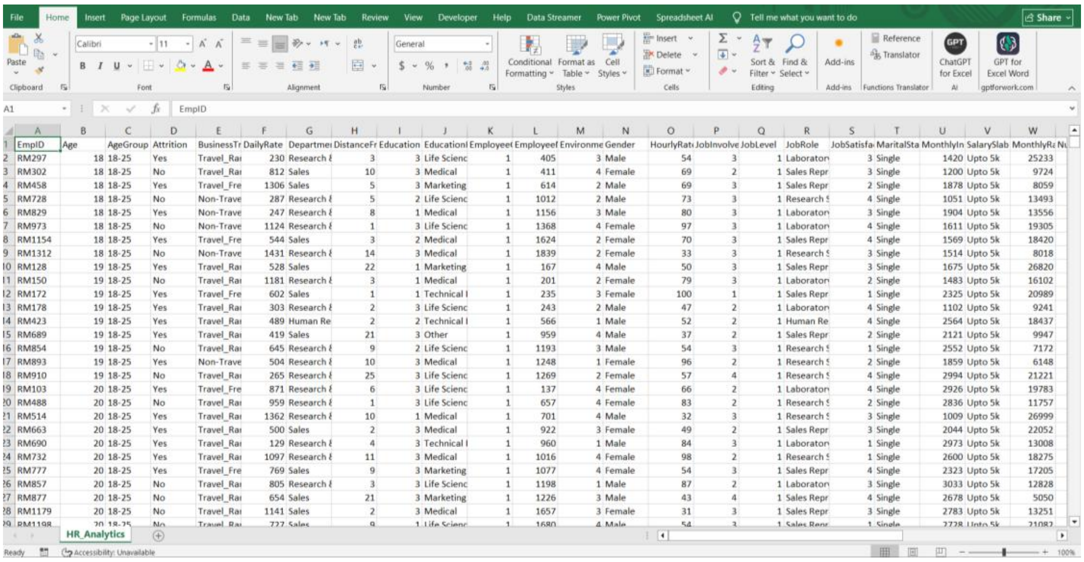


Summary

```
> summary(pca)
Importance of components:
                          PC1    PC2     PC3     PC4
Standard deviation     1.7084 0.9560 0.38309 0.14393
Proportion of Variance 0.7296 0.2285 0.03669 0.00518
Cumulative Proportion  0.7296 0.9581 0.99482 1.00000
```

Practical 10
Calculate Summary Statistics in Excel

Descriptive Statistics

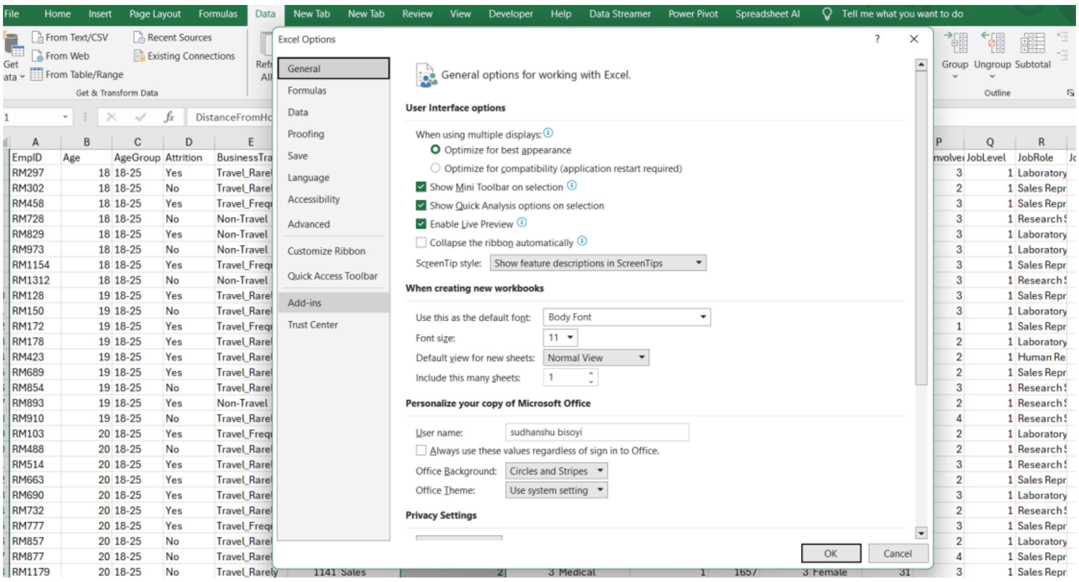


1.Inbuild Function

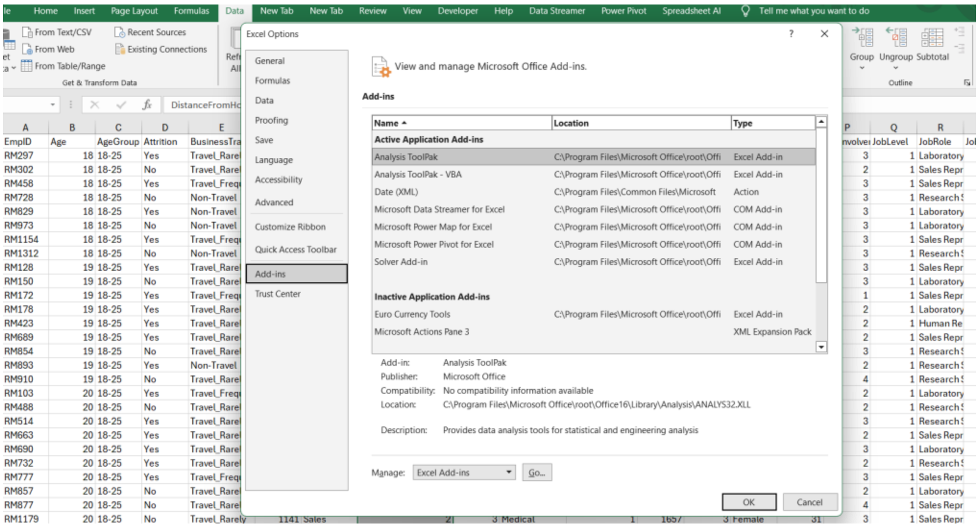
Average	=AVERAGE(H2:H1481)	9.22027027
Median	=MEDIAN(H2:H1481)	7
Mode	=MODE.SNGL(H2:H1481)	2
Standardd Deviation	=STDEV.P(H2:H1481)	8.128453191
Variance	=VAR.P(H2:H1481)	66.07175128
First Quartile	=QUARTILE.EXC(H2:H1481,1)	2
Third quartile	=QUARTILE.EXC(H2:H1481,3)	14
Range	=MAX(H2:H1481)-MIN(H2:H1481)	28
Minimun	=MIN(H2:H1481)	1
Maximun	=MAX(H2:H1481)	29
Sum	=SUM(H2:H1481)	13646
Count	=COUNT(H2:H1481)	1480

2. Tool Pak

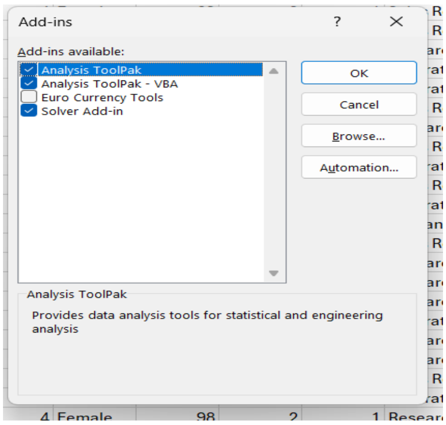
Click on File which is on top-left corner. then click on Options which is on bottom-left corner.

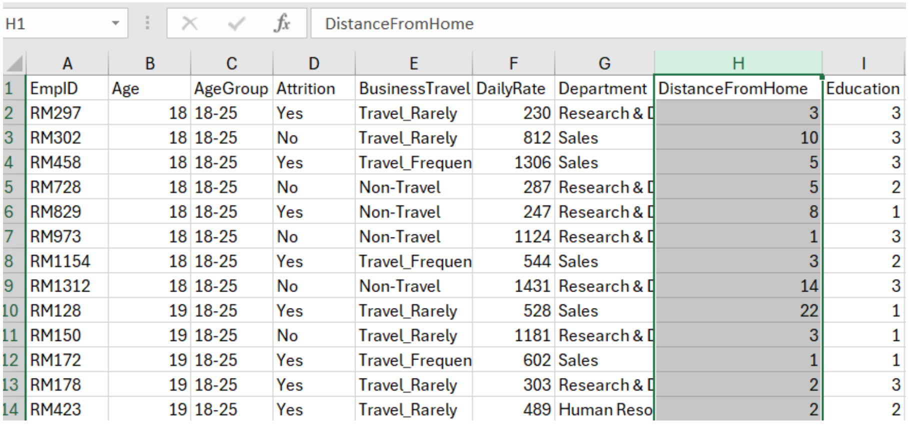


Then click on Add-ins option. Then Click on Go..

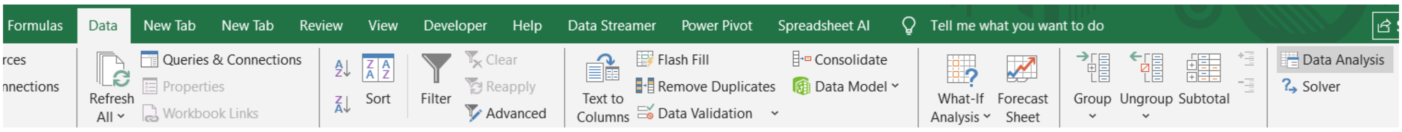


We will see Add-ins options and then check in 3 options and click on OK button.



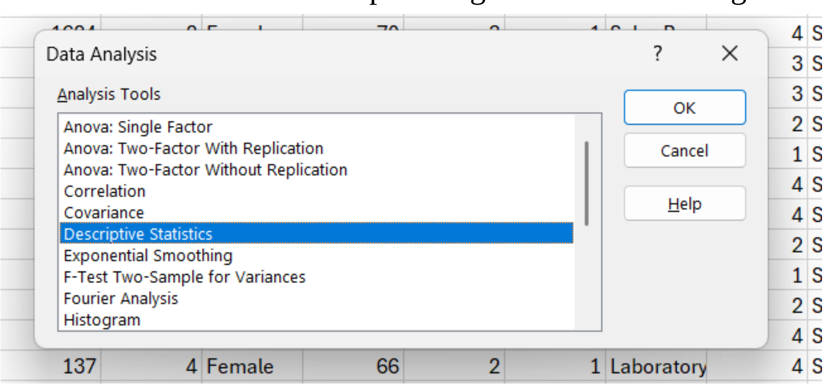


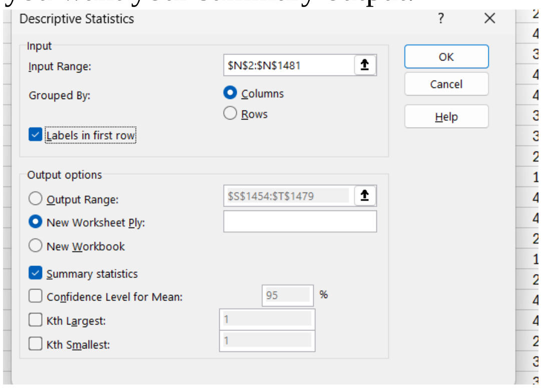
Click on Data which is located in ribbon at the top of the excel window. And then click on Data Analysis which is at the top-right corner.



Select Descriptive Statistics and press OK. Then select the input range from the column which we want to summarize. Check-in options which shown below.

If you want summary in new sheet then check-in new worksheet option or else you want on existing worksheet then check-in Output Range and select the range where you want your summary output.



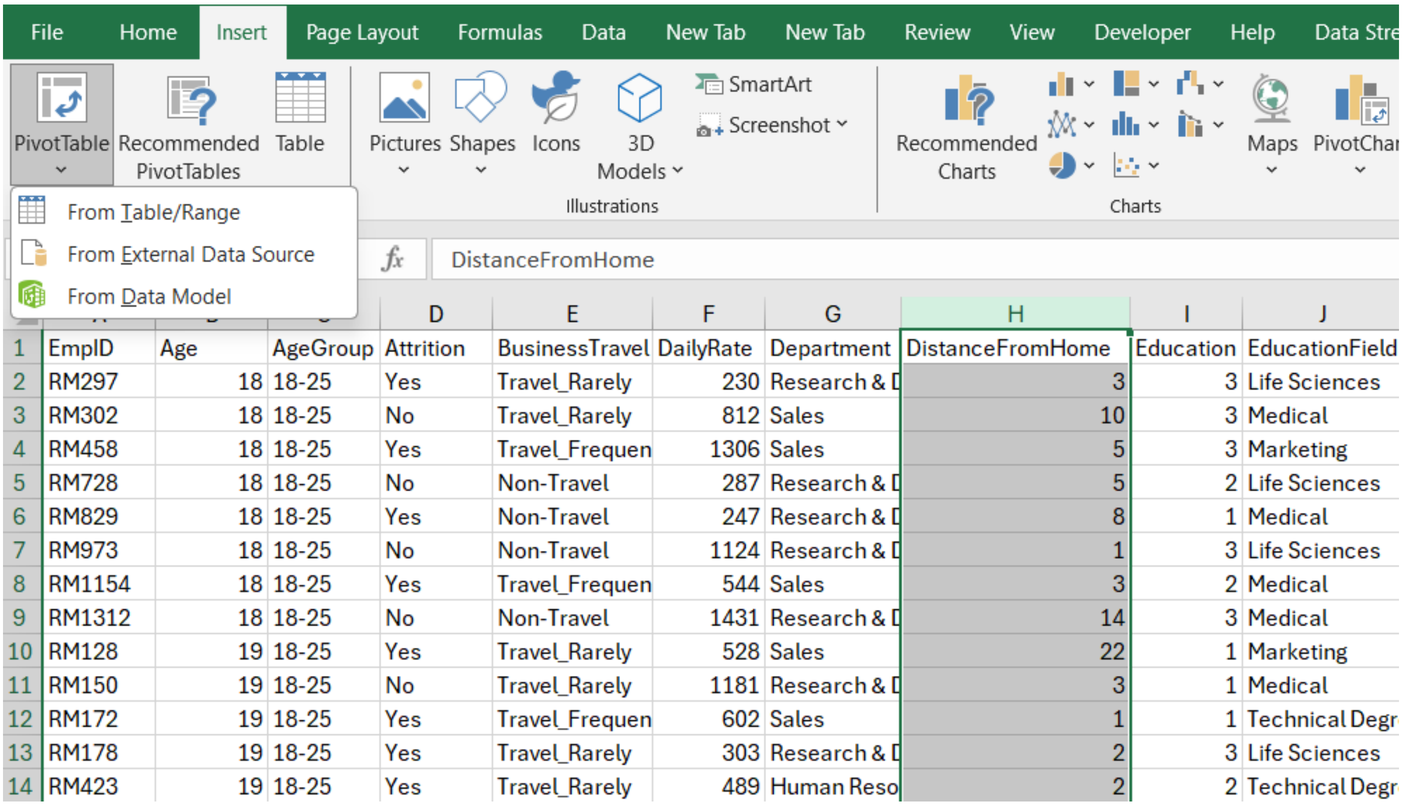


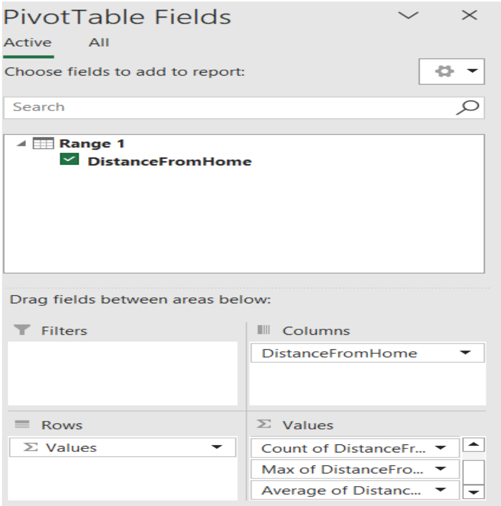
Summary Output:

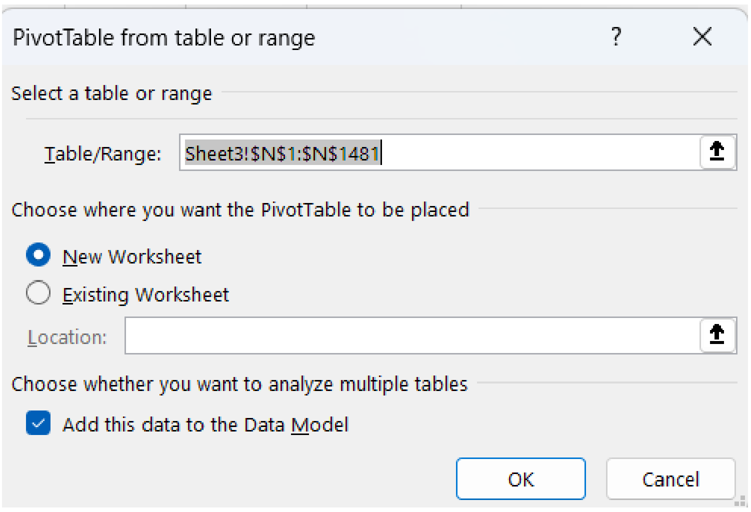
Summary	
Mean	9.22027027
Standard Error	0.211360497
Median	7
Mode	2
Standard Deviation	8.131200682
Sample Variance	66.11642454
Kurtosis	-0.237141939
Skewness	0.956145219
Range	28
Minimum	1
Maximum	29
Sum	13646
Count	1480

3.Pivot Table

In this case we want to pivot the column named [Distance From Home]. Then clicked on Insert from the ribbon toolbar and click on Pivot Table option then select [From Table/Range].







For the row wise pivot table you drag the values to Rows.

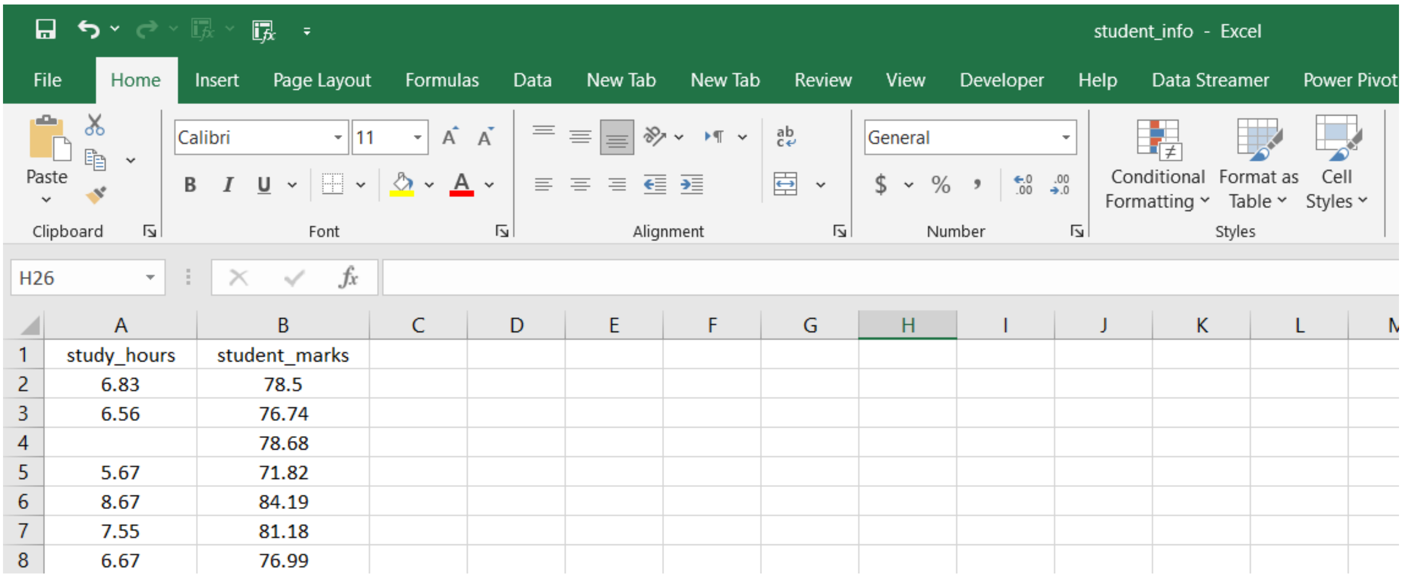
Values	
Max of DistanceFromHome	29
Sum of DistanceFromHome2	13646
StdDev of DistanceFromHome2	8.131200682
Count of DistanceFromHome2_2	1480
Average of DistanceFromHome2	9.22027027

For the column wise pivot table drag the values to Columns.

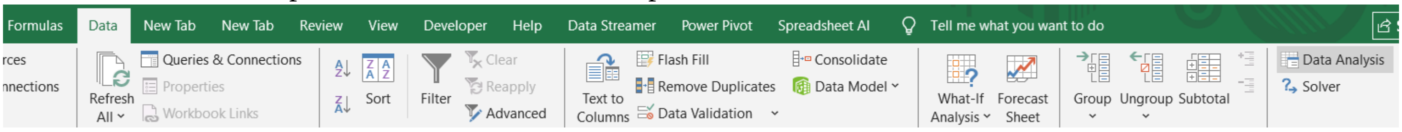
Row Labels	Sum of DistanceFromHome	Count of DistanceFromHome2	Max of DistanceFromHome2_2	Average of DistanceFromHome2
1	208	208	1	1
2	424	212	2	2
3	252	84	3	3
4	260	65	4	4
5	335	67	5	5
6	360	60	6	6
7	588	84	7	7
8	648	81	8	8
9	765	85	9	9
10	860	86	10	10
Grand Total	**4700**	**1032**	**10**	**4.554263566**

Practical 11
Generate Comparative Statistics in Excel

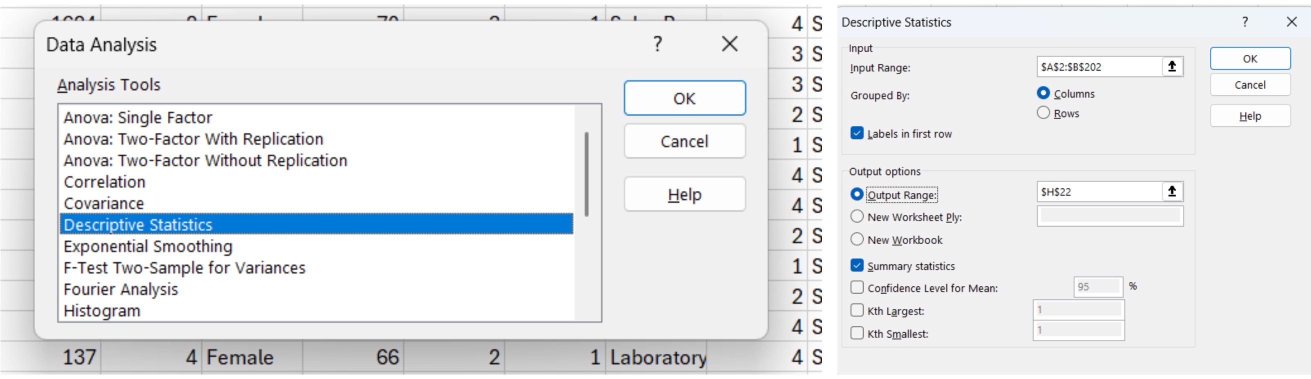
1.Two Numerical Column



Click on File which is on top-left corner. then click on Options which is on bottom-left corner.
Then click on Add-ins option. Then Click on Go..
We will see Add-ins options and then check in 3 options and click on OK button.



Select Descriptive Statistics and press OK. Then select the input range from the column which we want to summarize. Check-in options which shown below.



Comparative Statistics:

study hours		student marks	
Mean	6.995948718	Mean	77.93375
Standard Error	0.089733488	Standard Error	0.348299562
Median	7.12	Median	77.71
Mode	5.39	Mode	83.08
Standard Deviation	1.253059965	Standard Deviation	4.925699648
Sample Variance	1.570159276	Sample Variance	24.26251702
Kurtosis	-1.367950618	Kurtosis	-1.228159922
Skewness	-0.061310422	Skewness	-0.016924556
Range	3.98	Range	18.42
Minimum	5.01	Minimum	68.57
Maximum	8.99	Maximum	86.99
Sum	1364.21	Sum	15586.75
Count	195	Count	200

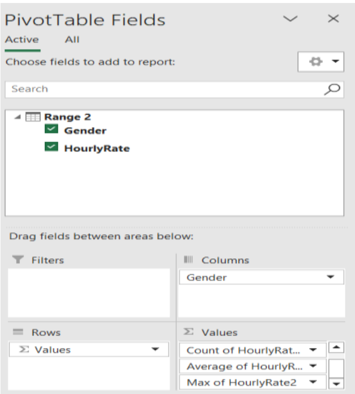
2.From HR_Analytics dataset we selected the 1 categorical and 1 numerical column

E	F	G	H	I	J	K	L	M	N	O	P	Q
essTravel	DailyRate	Departmer	DistanceFromHo	Education	EducationField	EmployeeCount	EmployeeN	Environme	Gender	HourlyRate	JobInvolve	JobLevel
_Rarely	230	Research &	3	3	Life Sciences	1	405	3	Male	54	3	
_Rarely	812	Sales	10	3	Medical	1	411	4	Female	69	2	
_Frequen	1306	Sales	5	3	Marketing	1	614	2	Male	69	3	
'ravel	287	Research &	5	2	Life Sciences	1	1012	2	Male	73	3	
'ravel	247	Research &	8	1	Medical	1	1156	3	Male	80	3	
'ravel	1124	Research &	1	3	Life Sciences	1	1368	4	Female	97	3	
_Frequen	544	Sales	3	2	Medical	1	1624	2	Female	70	3	
'ravel	1431	Research &	14	3	Medical	1	1839	2	Female	33	3	

ToolPak

Male Hourly Rate		Female Hourly Rate	
Mean	65.37394247	Mean	65.84433164
Standard Error	0.819116783	Standard Error	0.846722875
Median	65	Median	67
Mode	87	Mode	43
Standard Deviation	19.91313163	Standard Deviation	20.58424931
Sample Variance	396.5328114	Sample Variance	423.7113195
Kurtosis	-1.18335911	Kurtosis	-1.186529366
Skewness	0.005368015	Skewness	-0.0659771
Range	70	Range	70
Minimum	30	Minimum	30
Maximum	100	Maximum	100
Sum	38636	Sum	38914
Count	591	Count	591

3.Pivot Table



	Column Labels		
Values	Female	Male	Grand Total
Sum of HourlyRate	38914	58537	97451
Count of HourlyRate2	591	889	1480
Average of HourlyRat	65.84433164	65.84589426	65.84527027
Max of HourlyRate2	100	100	100

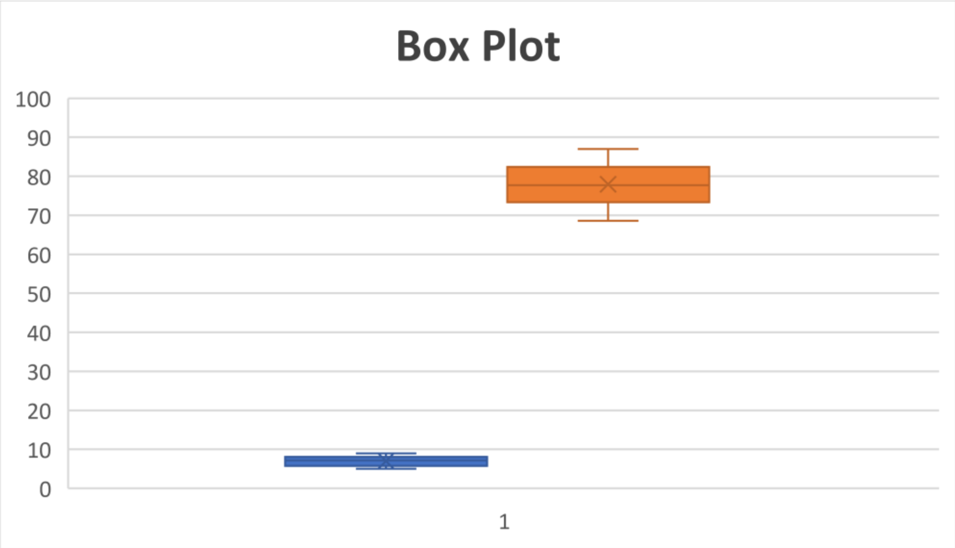
IF In-build

Mean	65.37394247
Standard Error	0.819116783
Median	65
Mode	87
Standard Deviation	19.91313163
Sample Variance	396.5328114
Kurtosis	-1.18335911
Skewness	0.005368015
Range	70
Minimum	30
Maximum	100
Sum	38636
Count	591

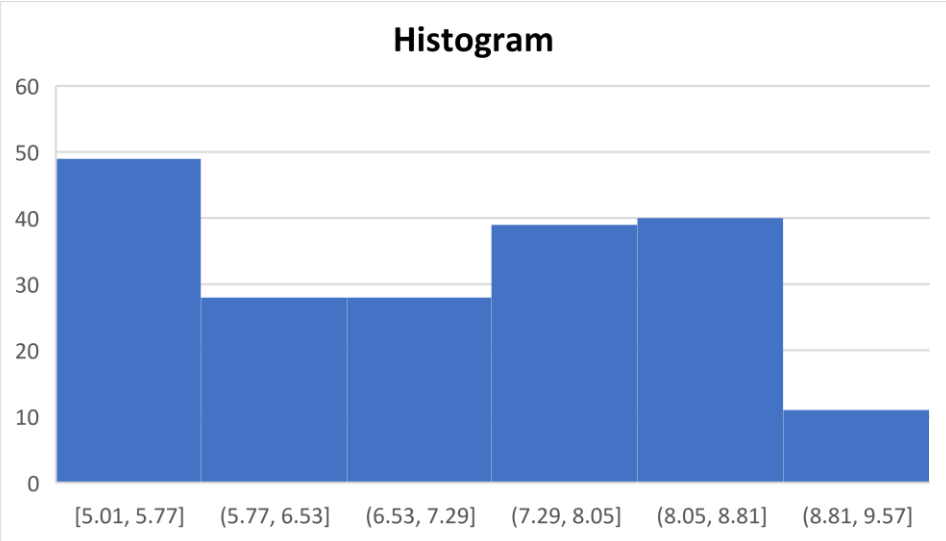
Practical 12
Create Graph in Excel

1. For 1 Numerical Column [Using Student_info dataset]

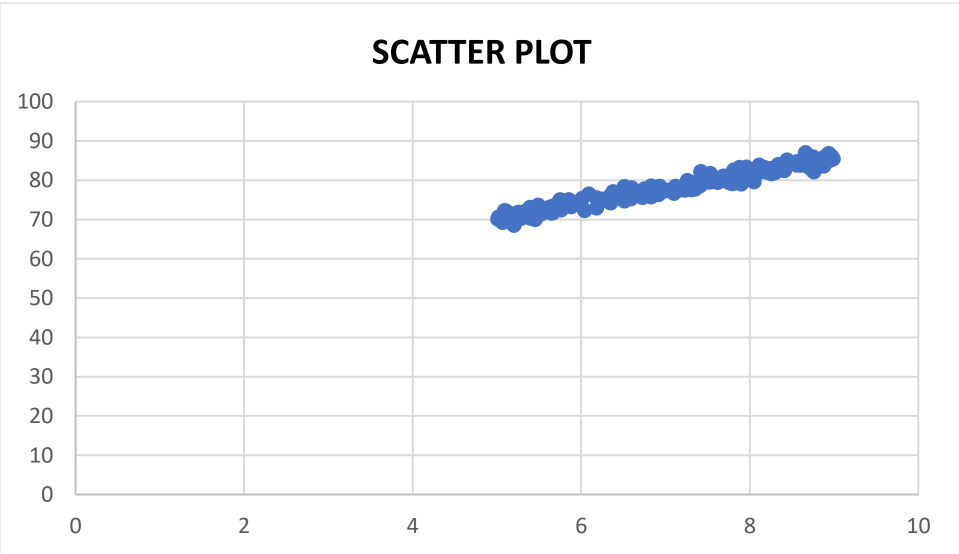
Box Plot



Histogram



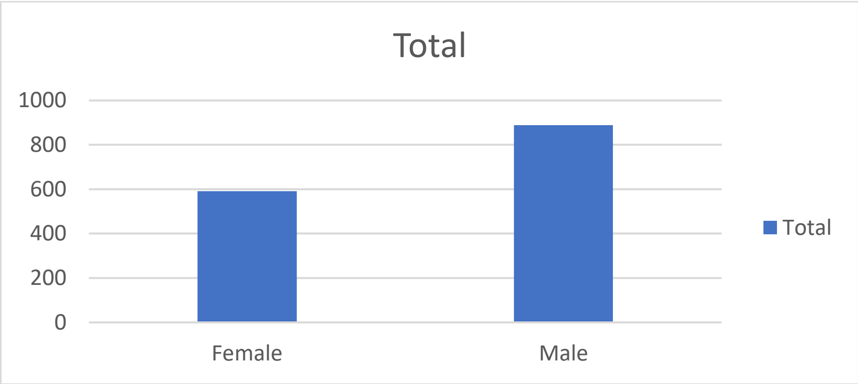
2. For Two Numerical Column [Using Student_info dataset]



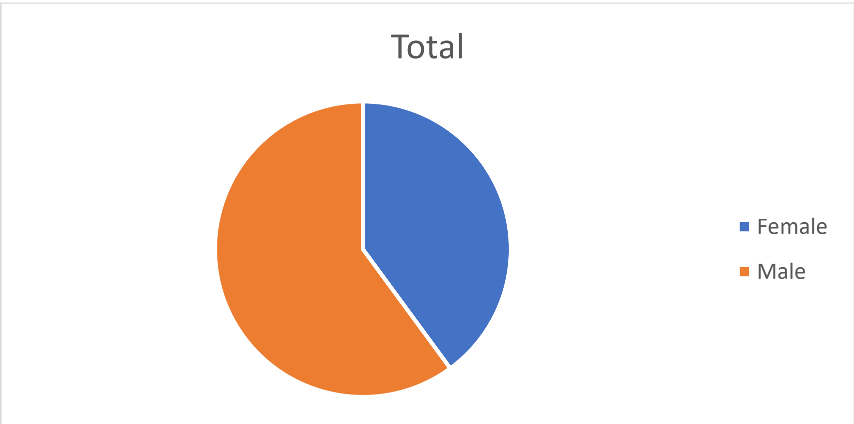
3. For 1 Categorical and 1 Numerical [Using HR_Analytics dataset]

Row Labels	Count of Gender
Female	591
Male	889
Grand Total	**1480**

Column Chart



Pie Chart



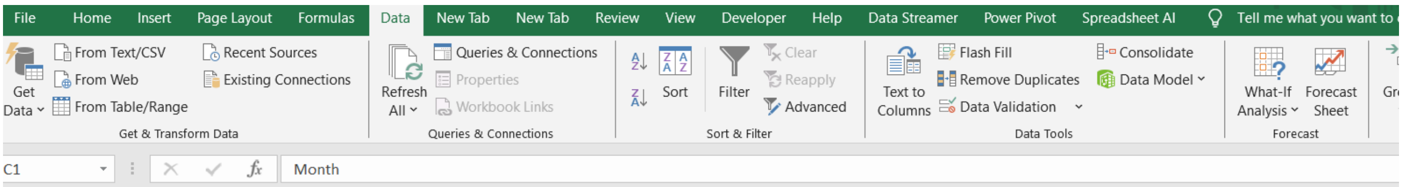
Month-Wise data

Using Sales data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Day	Month	Year	Customer_	Age_Grour	Customer_	Country	State	Product_C	Sub_Categ	Product	Order_Qua	Unit_Cost	Unit_Price	Profit	Cost	Revenue
2	11/26/2013	26	November	2013	19	Youth (<25	M	Canada	British Colı	Accessorie	Bike Racks	Hitch Rack	8	45	120	590	360	950
3	11/26/2015	26	November	2015	19	Youth (<25	M	Canada	British Colı	Accessorie	Bike Racks	Hitch Rack	8	45	120	590	360	950
4	3/23/2014	23	March	2014	49	Adults (35-	M	Australia	New South	Accessorie	Bike Racks	Hitch Rack	23	45	120	1366	1035	2401
5	3/23/2016	23	March	2016	49	Adults (35-	M	Australia	New South	Accessorie	Bike Racks	Hitch Rack	20	45	120	1188	900	2088
6	5/15/2014	15	May	2014	47	Adults (35-	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	4	45	120	238	180	418

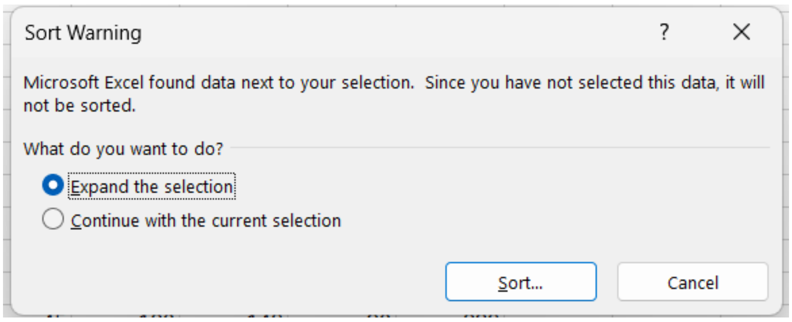
Sorted month-wise

Firstly, select the column you want to sort then click on Data which located in ribbon at the top-window.

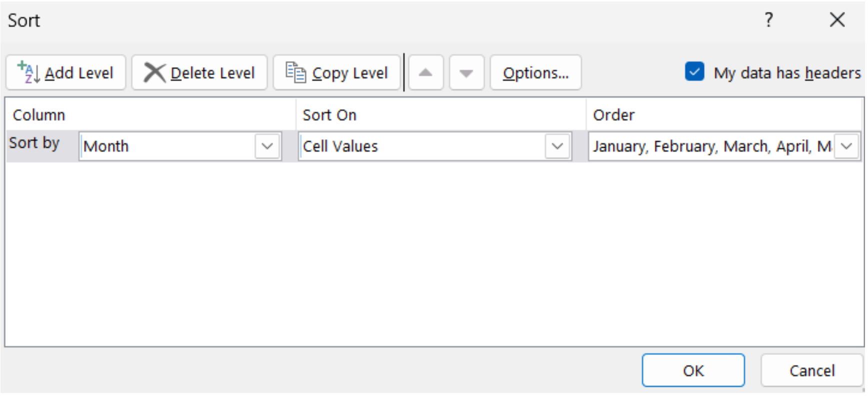


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Day	Month	Year	Customer_	Age_Grour	Customer_	Country	State	Product_C	Sub_Categ	Product	Order_Qua	Unit_Cost	Unit_Price	Profit	Cost	Revenue
2	11/26/2013	26	November	2013	19	Youth (<25	M	Canada	British Colı	Accessorie	Bike Racks	Hitch Rack	8	45	120	590	360	950
3	11/26/2015	26	November	2015	19	Youth (<25	M	Canada	British Colı	Accessorie	Bike Racks	Hitch Rack	8	45	120	590	360	950
4	3/23/2014	23	March	2014	49	Adults (35-	M	Australia	New South	Accessorie	Bike Racks	Hitch Rack	23	45	120	1366	1035	2401
5	3/23/2016	23	March	2016	49	Adults (35-	M	Australia	New South	Accessorie	Bike Racks	Hitch Rack	20	45	120	1188	900	2088
6	5/15/2014	15	May	2014	47	Adults (35-	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	4	45	120	238	180	418

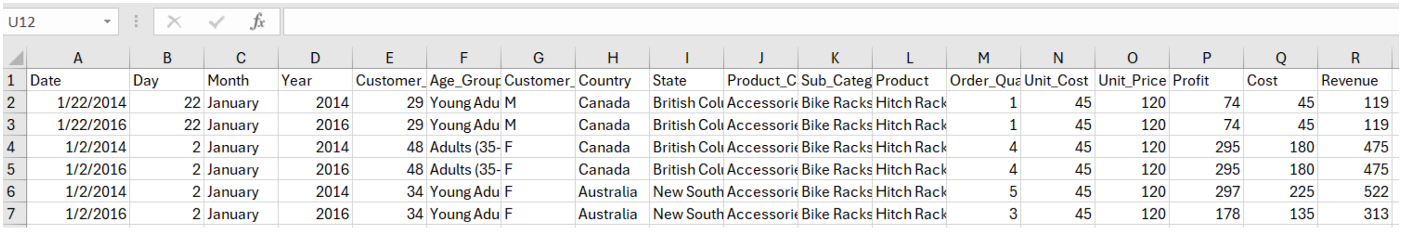
Click on Sort which is at below the view toolbar.



Then choose the option Month in sort by and by choosing custom select the order by month.



Now, it is sorted by month

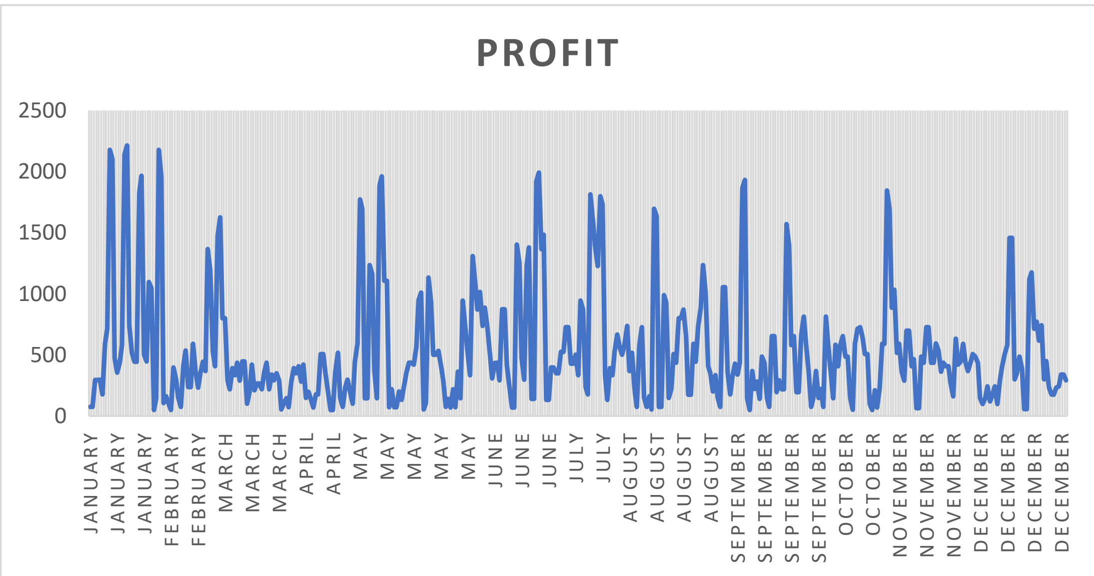


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Day	Month	Year	Customer_	Age_Grour	Customer_	Country	State	Product_C	Sub_Categ	Product	Order_Qua	Unit_Cost	Unit_Price	Profit	Cost	Revenue
2	1/22/2014	22	January	2014	29	Young Adu	M	Canada	British Colı	Accessorie	Bike Racks	Hitch Rack	1	45	120	74	45	119
3	1/22/2016	22	January	2016	29	Young Adu	M	Canada	British Colı	Accessorie	Bike Racks	Hitch Rack	1	45	120	74	45	119
4	1/2/2014	2	January	2014	48	Adults (35-	F	Canada	British Colı	Accessorie	Bike Racks	Hitch Rack	4	45	120	295	180	475
5	1/2/2016	2	January	2016	48	Adults (35-	F	Canada	British Colı	Accessorie	Bike Racks	Hitch Rack	4	45	120	295	180	475
6	1/2/2014	2	January	2014	34	Young Adu	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	5	45	120	297	225	522
7	1/2/2016	2	January	2016	34	Young Adu	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	3	45	120	178	135	313

Line Graph With Pivot Table

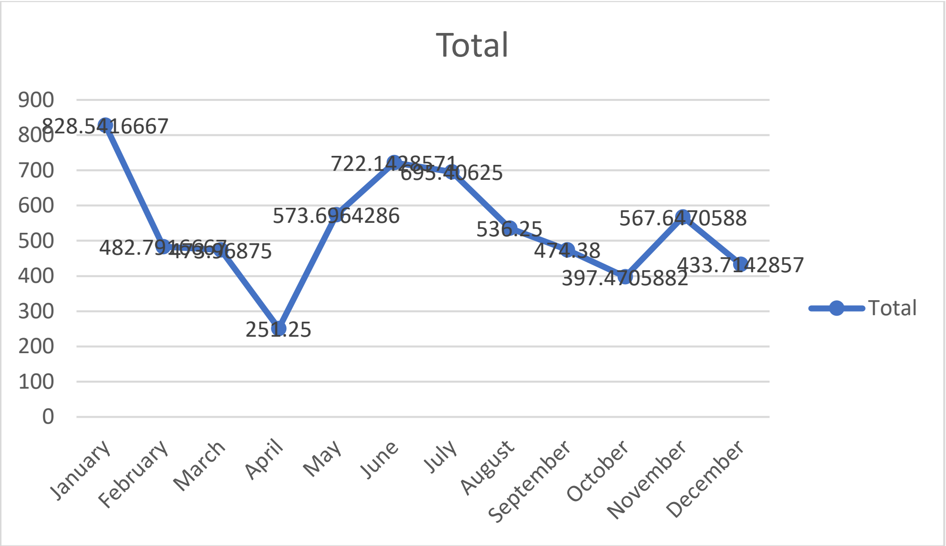
Select the columns and click on insert and then select line chart on chart section.

Month	Profit
January	74
January	74
January	295
January	295
January	297
January	178
January	594
January	713
January	2178
January	2105
January	475
January	356
January	436
January	581



Line Graph Without Pivot Table

Row Labels	Average of Profit
January	828.5416667
February	482.7916667
March	473.96875
April	251.25
May	573.6964286
June	722.1428571
July	695.40625
August	536.25
September	474.38
October	397.4705882
November	567.6470588
December	433.7142857
Grand Total	**537.7593985**



Practical 13
Advanced Data Analysis using PivotTables and Pivot Charts

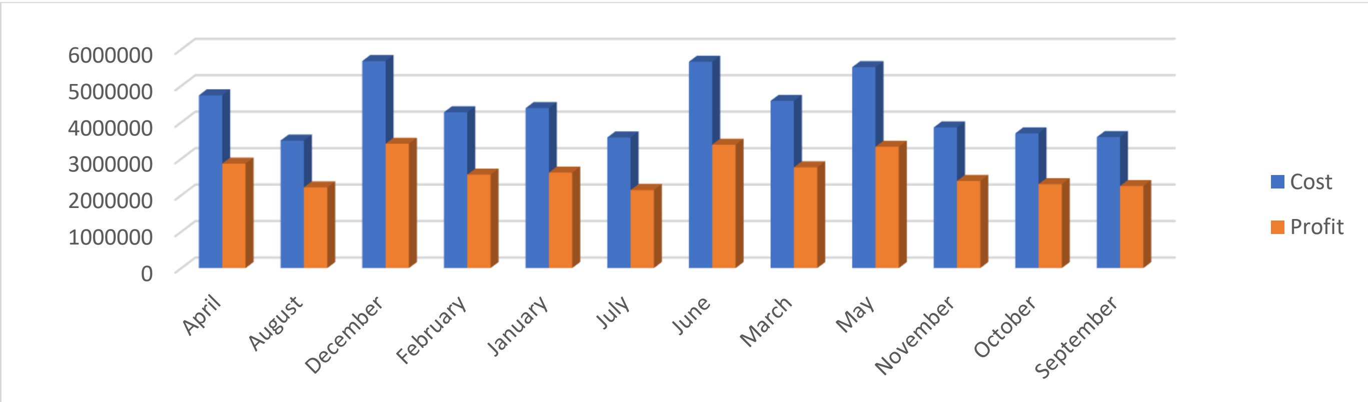
Using Sales Data

U12

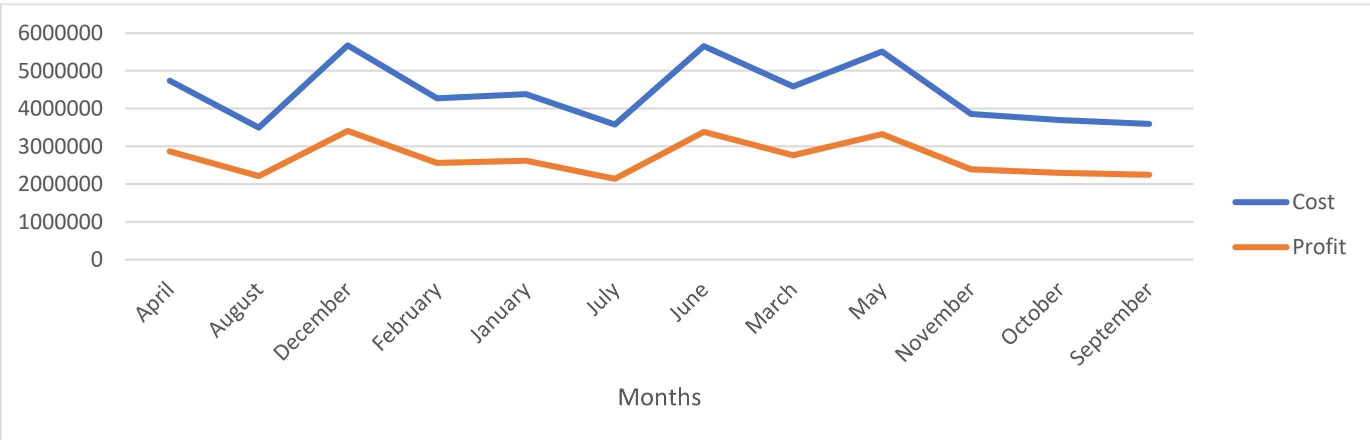
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Day	Month	Year	Customer_	Age_Group	Customer_	Country	State	Product_C	Sub_Categ	Product	Order_Qua	Unit_Cost	Unit_Price	Profit	Cost	Revenue
2	1/22/2014	22	January	2014	29	Young Adu	M	Canada	British Col	Accessorie	Bike Racks	Hitch Rack	1	45	120	74	45	119
3	1/22/2016	22	January	2016	29	Young Adu	M	Canada	British Col	Accessorie	Bike Racks	Hitch Rack	1	45	120	74	45	119
4	1/2/2014	2	January	2014	48	Adults (35-	F	Canada	British Col	Accessorie	Bike Racks	Hitch Rack	4	45	120	295	180	475
5	1/2/2016	2	January	2016	48	Adults (35-	F	Canada	British Col	Accessorie	Bike Racks	Hitch Rack	4	45	120	295	180	475
6	1/2/2014	2	January	2014	34	Young Adu	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	5	45	120	297	225	522
7	1/2/2016	2	January	2016	34	Young Adu	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	3	45	120	178	135	313

Row Labels	Cost	Profit
April	4738031	2864719
August	3496989	2214204
December	5677219	3409712
February	4272261	2562322
January	4387374	2618521
July	3581709	2139750
June	5659425	3383583
March	4585298	2761866
May	5509826	3326937
November	3855785	2388513
October	3693767	2301312
September	3592224	2249661
Grand Total	**53049908**	**32221100**

Column Chart



Line Chart



Practical 14

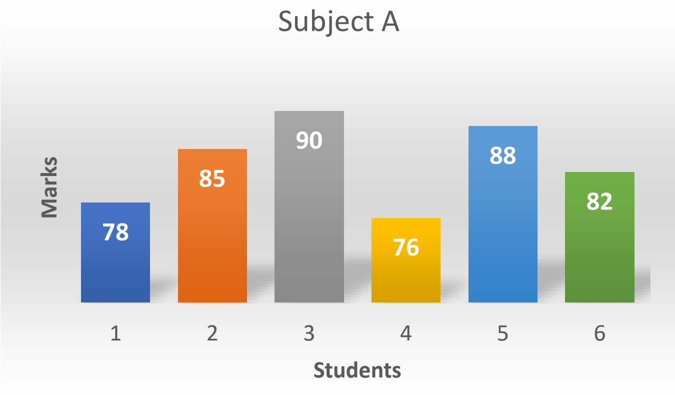
Tabulation, bar diagram, Multiple Bar diagram, Pie diagram, Measure of central tendency: Mean, median, mode, Measure of dispersion: variance, standard deviation, Coefficient of variation. Correlation, regression lines

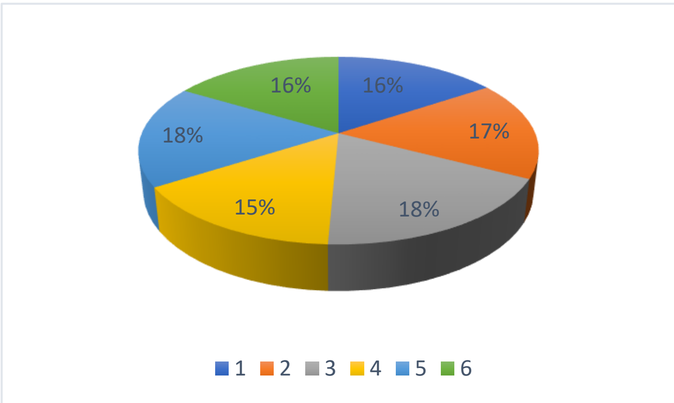
Student	Subject A	Subject B
S1	78	65
S2	85	70
S3	90	80
S4	76	60
S5	88	75
S6	82	68

For Subject A

Measure of Central Tendency		
Mean	83.16667	69.66667
Median	83.5	
Mode	No Mode	

Measures of Dispersion	
Variance	30.56666667
Standard Deviation	5.04700131
Coefficient of Variation	6.647751019

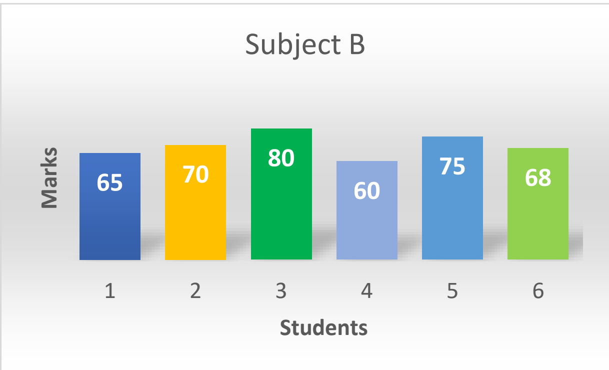


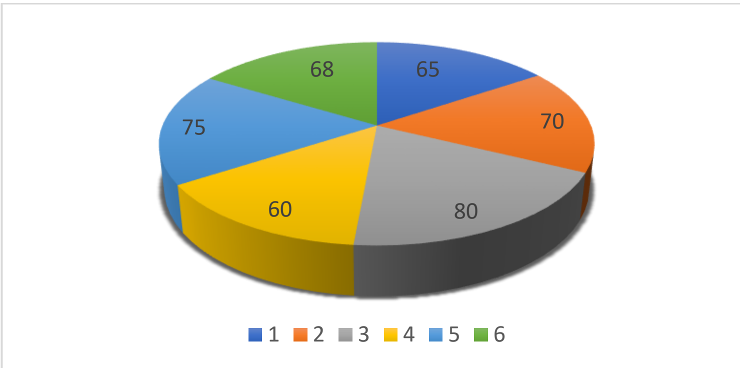


For Subject B

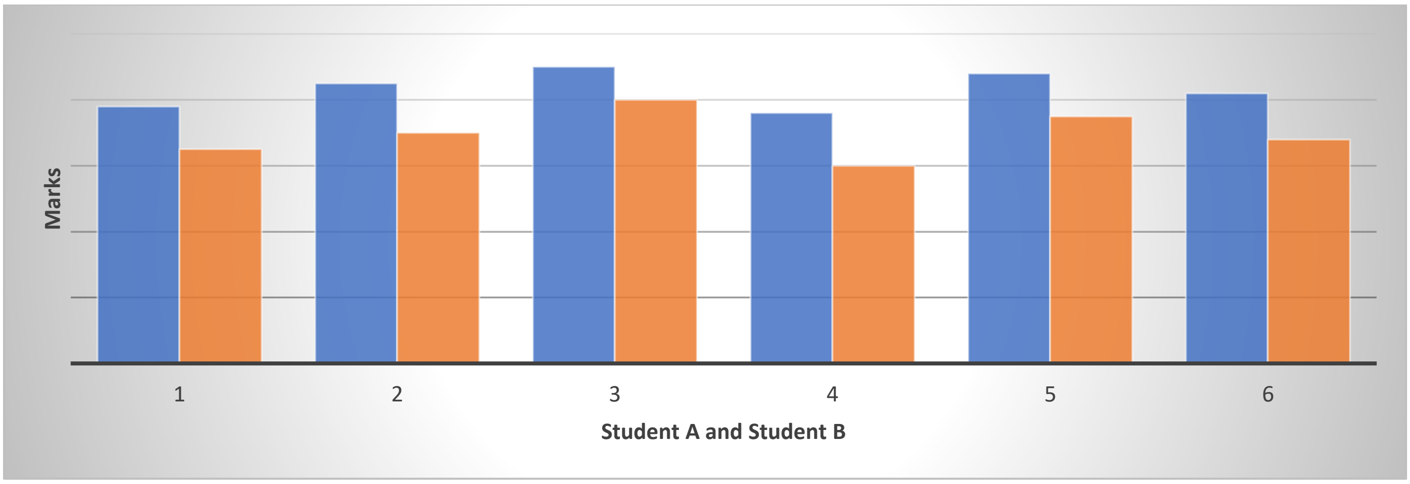
Measure of Central Tendency	
Mean	69.66666667
Median	69
Mode	No Mode

Measures of Dispersion	
Variance	50.66666667
standard deviation	6.497862897
coefficient of variation	10.21729976

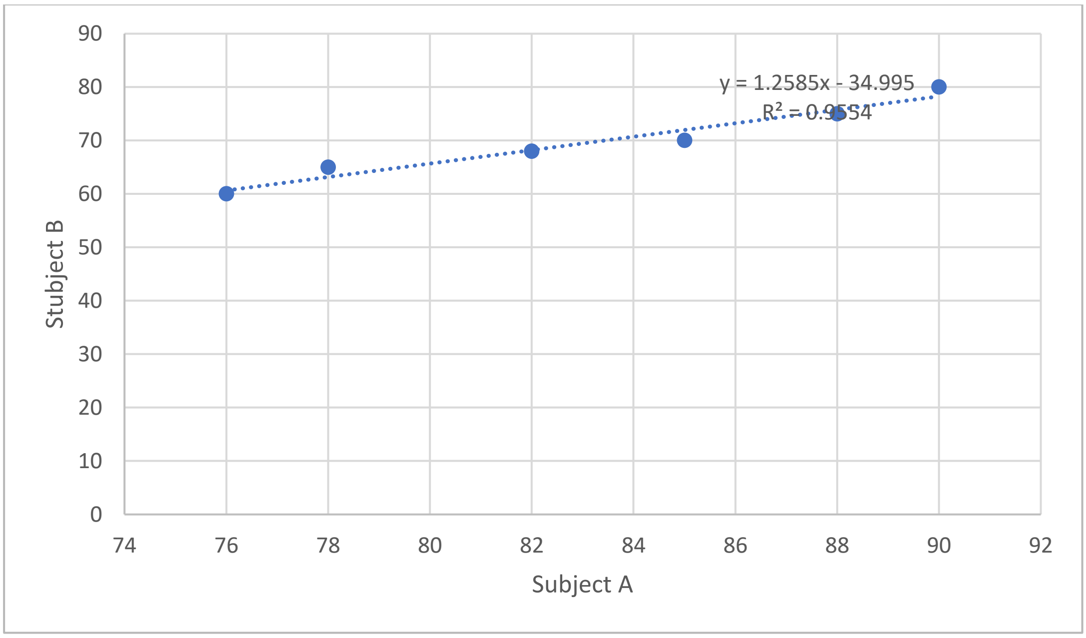




Multiple Bar Plot



Scatter Plot



Practical 15

t-test , F-test, ANOVA one way classification, chi square test, independence of attributes

t-test

t-Test: Two-Sample Assuming Unequal Variances

Boys Marks	Girls Marks
78	70
82	75
85	72
80	74
76	71

	Boys Marks	*Girls Marks*
Mean	80.2	72.4
Variance	12.2	4.3
Observations	5	5
Hypothesized Mean Difference	0	
df	7	
t Stat	4.2937586839928	=IF(G11<F13,"Reject H0","Accept H0")
P(T<=t) one-tail	0.0017970245301787	=IF(G12<0.05,"Reject H0","Accept H0")
t Critical one-tail	1.89457860509001	=IF(G11<F13,"Reject H0","Accept H0")
P(T<=t) two-tail	0.0035940490603574	=IF(G14<0.05,"Reject H0","Accept H0")
t Critical two-tail	2.36462425159278	=IF(G11<F13,"Reject H0","Accept H0")

Girls Marks
72.4
4.3
5
Reject H0
Reject H0
Reject H0
Reject H0
Reject H0

F-test

F-Test Two-Sample for Variances

Machine A	Machine B
12	10
14	11
15	9
13	10
16	12

	Machine A	*Machine B*
Mean	14	10.4
Variance	2.5	1.3
Observations	5	5
df	4	4
F	1.92307692307692	=IF(G25<G27,"Accept H0","Reject H0")
P(F<=f) one-tail	0.271030762501822	=IF(G26<0.05,"Reject H0","Accept H0")
F Critical one-tail	6.38823290869587	=IF(G25<G27,"Accept H0","Reject H0")

Machine B
10.4
1.3
5
4
Accept H0
Accept H0
Accept H0

Conclusion: The calculated value is less than critical value so we accept the Null Hypothesis.

ANOVA one way classification

Anova: Single Factor

Region A	Region B	Region C
52	60	55
49	62	53
50	58	56
51	61	54

Summary

Groups	*Count*	*Sum*	*Average*	*Variance*
Region A	4	202	50.5	1.666666667
Region B	4	241	60.25	2.916666667
Region C	4	218	54.5	1.666666667

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	192.16667	2	96.08333	46.12	1.86E-05	4.256495
Within Groups	18.75	9	2.083333			
Total	210.91667	11				

Conclusion – we reject null hypothesis

Chi-Square test

	Product A	Product B	Product C	Total
Male	20	15	25	60
Female	30	25	20	75
Total	50	40	45	135

	Product A	Product B	Product C	Total
Male	22.222222	17.77777778	20	60
Female	27.777778	22.22222222	25	75
Total	50	40	45	135

Chi-Squre test	**=CHITEST(B51:E53,B57:E59)**	**0.75309298**

Conclusion – we reject null hypothesis

Practical 16

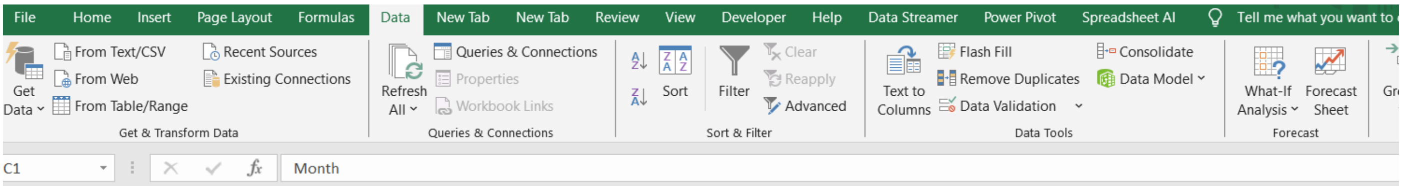
Time series: forecasting Method of least squares, moving average method. Inference and discussion of results

Using Sales data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Day	Month	Year	Customer_	Age_Grour	Customer_	Country	State	Product_C	Sub_Categ	Product	Order_Qua	Unit_Cost	Unit_Price	Profit	Cost	Revenue
2	11/26/2013	26	November	2013	19	Youth (<25	M	Canada	British Colu	Accessorie	Bike Racks	Hitch Rack	8	45	120	590	360	950
3	11/26/2015	26	November	2015	19	Youth (<25	M	Canada	British Colu	Accessorie	Bike Racks	Hitch Rack	8	45	120	590	360	950
4	3/23/2014	23	March	2014	49	Adults (35-	M	Australia	New South	Accessorie	Bike Racks	Hitch Rack	23	45	120	1366	1035	2401
5	3/23/2016	23	March	2016	49	Adults (35-	M	Australia	New South	Accessorie	Bike Racks	Hitch Rack	20	45	120	1188	900	2088
6	5/15/2014	15	May	2014	47	Adults (35-	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	4	45	120	238	180	418

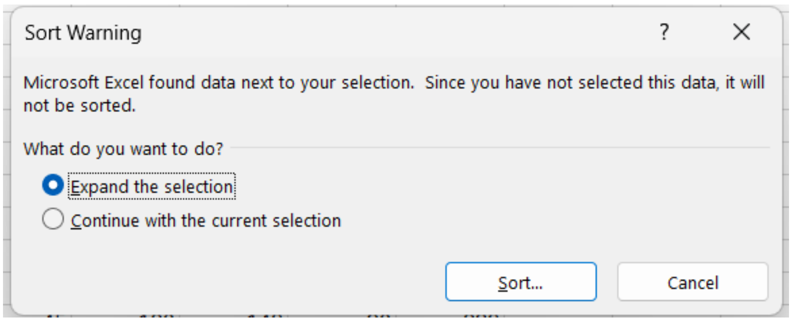
Sorted month-wise

Firstly, select the column you want to sort then click on Data which located in ribbon at the top-window.

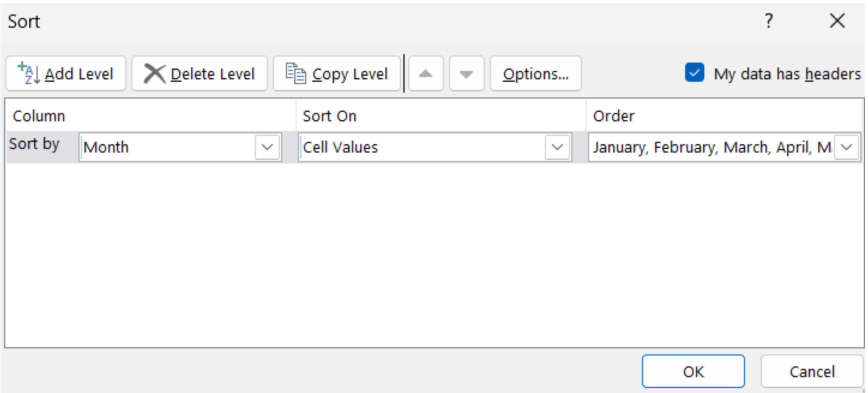


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Day	Month	Year	Customer_	Age_Grour	Customer_	Country	State	Product_C	Sub_Categ	Product	Order_Qua	Unit_Cost	Unit_Price	Profit	Cost	Revenue
2	11/26/2013	26	November	2013	19	Youth (<25	M	Canada	British Colu	Accessorie	Bike Racks	Hitch Rack	8	45	120	590	360	950
3	11/26/2015	26	November	2015	19	Youth (<25	M	Canada	British Colu	Accessorie	Bike Racks	Hitch Rack	8	45	120	590	360	950
4	3/23/2014	23	March	2014	49	Adults (35-	M	Australia	New South	Accessorie	Bike Racks	Hitch Rack	23	45	120	1366	1035	2401
5	3/23/2016	23	March	2016	49	Adults (35-	M	Australia	New South	Accessorie	Bike Racks	Hitch Rack	20	45	120	1188	900	2088
6	5/15/2014	15	May	2014	47	Adults (35-	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	4	45	120	238	180	418

Click on Sort which is at below the view toolbar.



Then choose the option Month in sort by and by choosing custom select the order by month.



Now, it is sorted by month

U12

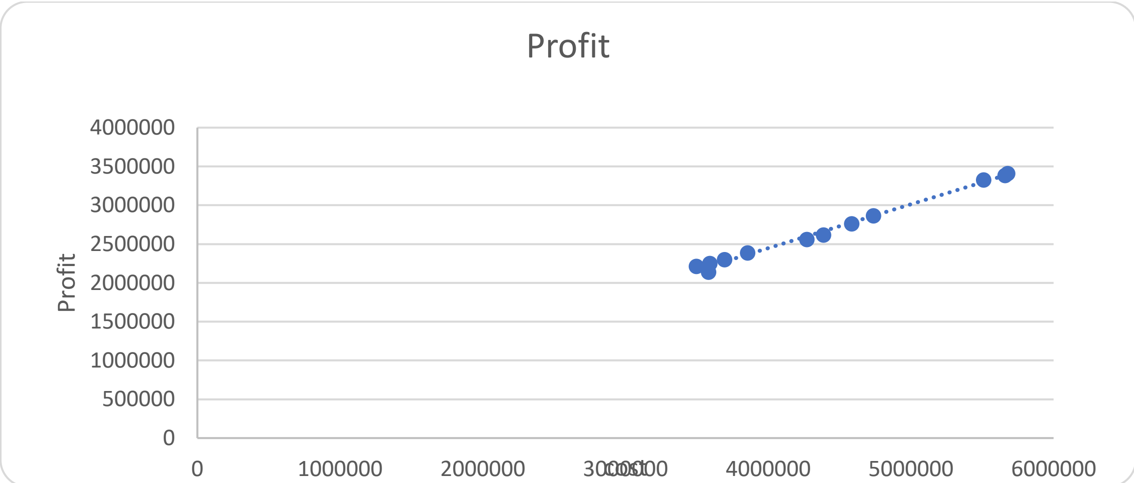
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	Day	Month	Year	Customer_	Age_Grour	Customer_	Country	State	Product_C	Sub_Categ	Product	Order_Qua	Unit_Cost	Unit_Price	Profit	Cost	Revenue
2	1/22/2014	22	January	2014	29	Young Adu	M	Canada	British Col	Accessorie	Bike Racks	Hitch Rack	1	45	120	74	45	119
3	1/22/2016	22	January	2016	29	Young Adu	M	Canada	British Col	Accessorie	Bike Racks	Hitch Rack	1	45	120	74	45	119
4	1/2/2014	2	January	2014	48	Adults (35-	F	Canada	British Col	Accessorie	Bike Racks	Hitch Rack	4	45	120	295	180	475
5	1/2/2016	2	January	2016	48	Adults (35-	F	Canada	British Col	Accessorie	Bike Racks	Hitch Rack	4	45	120	295	180	475
6	1/2/2014	2	January	2014	34	Young Adu	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	5	45	120	297	225	522
7	1/2/2016	2	January	2016	34	Young Adu	F	Australia	New South	Accessorie	Bike Racks	Hitch Rack	3	45	120	178	135	313

Method of Least Squares

Months	Cost	Profit
January	4387374	2618521
February	4272261	2562322
March	4585298	2761866
April	4738031	2864719
May	5509826	3326937
June	5659425	3383583
July	3581709	2139750
August	3496989	2214204
September	3592224	2249661
October	3693767	2301312
November	3855785	2388513
December	5677219	3409712

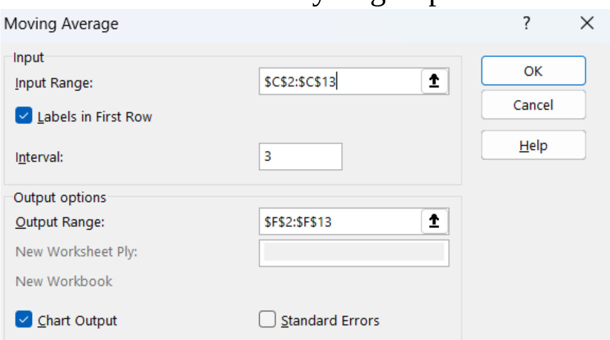
Slope	b	0.56549
Intercept	a	185159.5

y=a+bt
y=185159.5+0.56549



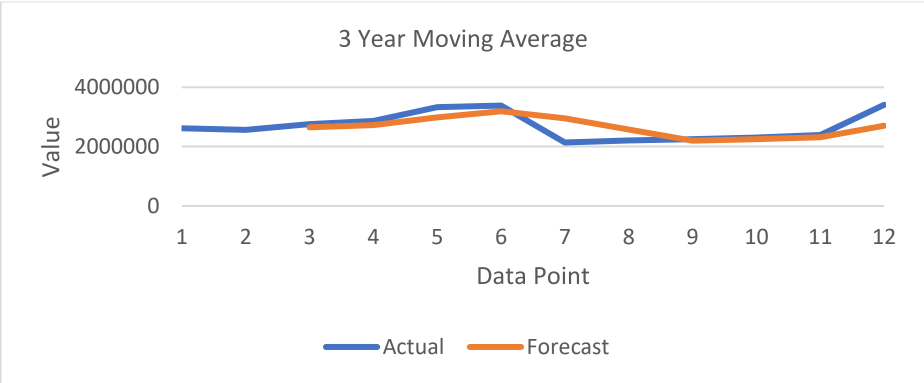
Moving Average Method

Ribbon → Data → Analysis group → Data Analysis → Moving Average → Ok

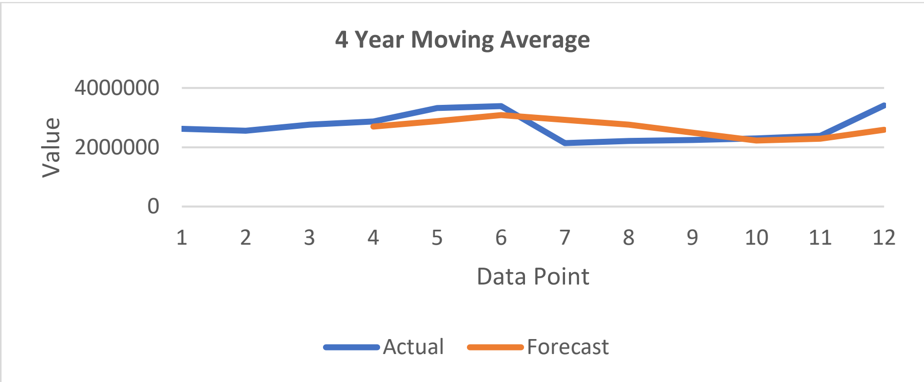


Months	Cost	Profit	3-year	4-year	5-year
January	4387374	2618521	#N/A	#N/A	#N/A
February	4272261	2562322	#N/A	#N/A	#N/A
March	4585298	2761866	2647570	#N/A	#N/A
April	4738031	2864719	2729636	2701857	#N/A
May	5509826	3326937	2984507	2878961	2979885
June	5659425	3383583	3191746	3084276	2895371
July	3581709	2139750	2950090	2928747	2785839
August	3496989	2214204	2579179	2766119	2662827
September	3592224	2249661	2201205	2496800	2457702
October	3693767	2301312	2255059	2226232	2258688
November	3855785	2388513	2313162	2288423	2512680
December	5677219	3409712	2699846	2587300	#N/A

3-Year Moving Average



4-Year Moving Average



5-Year Moving Average